

Optimal Portfolio Execution using Market and Limit Orders*

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Abstract

We investigate trade execution strategies that maximize expected exponential utility. In contrast to existing literature our strategy dynamically executes both market and limit orders. We derive a Hamilton-Jacobi-Bellman equation for the optimal combination of these order types and solve it numerically. We show that limit orders have a significant impact on the optimal strategy, and discuss the utility gains with respect to pure market order strategies. Our findings indicate an inverse relation between the speed at which market orders are submitted and the likelihood of limit orders being filled. For limit orders, we find that the optimal price moves away from the market price when any of the following hold: i) trade execution is not urgent, ii) the portfolio is relatively small, iii) the agent is not very risk averse. Finally, the optimal size of limit orders is largest for moderate portfolio sizes and long time horizons.

Keywords: Optimal trade execution, liquidity effects, market order, limit order, HJB equation, partial integro-differential equation, ADI method.

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Introduction

One of the assumptions often used in financial modeling is that assets can be traded without influencing prices. This assumption is not overly restrictive when individual trades represent only a tiny fraction of the market, but may not hold otherwise. Splitting large orders into gradually executed smaller ones may reduce the price impact, but in turn introduces exposure to price fluctuations. The solution hinges on finding the optimal balance between market risk and the execution costs related to price impact. In the last decade, a vast and growing literature revolved around the question of how to solve this trade-off in an optimal manner including, among others, [Bertsimas and Lo \[1998\]](#), [Almgren and Chriss \[2001\]](#), [Obizhaeva and Wang \[2005\]](#) and [Alfonsi et al. \[2010\]](#).

Existing literature considers two approaches to model the liquidity effect on prices. One approach is to capture them endogenously by modeling the mechanisms underlying liquidity. This leads to a general

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equilibrium analysis, developed in [Kyle \[1985\]](#) and [Easley and O'Hara \[1987\]](#), that is typically not being used for the analysis of optimal execution problems. A second approach takes the liquidity effects as given and solves the optimal execution problem in the resulting partial equilibrium model. Most studies on optimal execution take place within this framework. Here some authors, notably [Obizhaeva and Wang \[2005\]](#) and [Alfonsi et al. \[2010\]](#), assume the existence of a limit order book through which a market order can be executed. Others, such as [Bertsimas and Lo \[1998\]](#), [Almgren and Chriss \[2001\]](#) and [Almgren \[2003\]](#), assume that the effect of trades on prices can be described with a price impact function (PIF).

Research in the latter category can be further distinguished into continuous (see, for example, [Almgren \[2009\]](#)) and discrete approaches (see, for example, [Almgren and Chriss \[2001\]](#) and [Alfonsi et al. \[2010\]](#) for deterministic time steps or [Kharroubi and Pham \[2010\]](#) for random time steps). Although real-world trading is not done in a continuous setting, the formulation in continuous is typically accepted as a reasonable approximation of discrete trading given the growing prevalence of high-frequency trading.

The afore-mentioned literature is largely confined to the study of pure market order strategies. Only recently there has been an increased focus on considering a different, potentially broader, set of strategies. An interesting example is the paper of [Kratz and Schöneborn \[2012\]](#), who consider agents that can sell assets using market orders and a dark pool. [Esser and Monch \[2007\]](#), [Guéant et al. \[2012\]](#) and [Bayraktar and Ludkovski \[2012\]](#) analyze limit order strategies that only use market orders as a fall-back should there be shares left. While this approach leads to insightful results it doesn't constitute an extension of the original problem since market orders are no longer optimally distributed over time. One can therefore neither study the impact of limit orders on the agents utility nor draw conclusions regarding the optimal use of both order types simultaneously.

In this paper we extend the original problem of how to schedule orders in an optimal fashion through the combined use of market and limit orders. Compared to market orders, limit orders have the advantage of a negative liquidity cost since the order is executed on the other side of the market, but run the risk of not being filled quickly, or at all. This risk, which we will subsequently refer to as transaction risk, increases when the limit price moves further away from the market price. Thus it is generally beneficial to place limit orders close to the market price in order to minimize transaction risk. However, this will lead to a worse execution price. This consideration suggests the existence of an optimal limit price. The size of a limit order is subject to another trade-off. Posting a small-size limit order reduces the opportunity to earn, rather than pay, but since a limit order reveals a need to buy or sell, placing a large-size one may push the price away from the order. Striking the right balance will yield an optimal limit size, which, in addition to the optimal limit price and the optimal submission rate of market orders, is what we are looking to solve for.

The optimal use of market and limit orders has been studied in a number of different contexts. [Harris \[1998\]](#) discusses optimal order submission strategies for risk-neutral agents with heterogeneous information who may submit either a market or a limit order at a discrete set of points in time. [Avellaneda and Stoikov \[2008\]](#) and [Guilbaud and Pham \[2013\]](#) study optimal strategies for stock dealer's who provide liquidity by submitting bid and ask quotes in a limit order book. [Kühn and Stroh \[2010\]](#) extend Merton's portfolio optimization problem by assuming that agents may use limit orders to profit from the bid-ask spread. [Cont and Kukanov \[2012\]](#) concentrate on the issue of distributing a scheduled order across trading venues and order types.

The aim of this paper is twofold. First, to develop a mathematical framework for trade execution using a combination of market and limit orders. Second, to find the combination of market and limit orders that yields the ideal balance between market risk, transaction risk and execution costs.

Following [Almgren and Chriss \[2001\]](#), the impact of trading on prices is modeled with two PIFs, namely, a permanent PIF relating order flow imbalance to price changes, and a temporary PIF capturing the liquidity premium paid by market order submitters. The permanent PIF is based on some insights

from empirical studies on the price impact of order book events (see [Cont et al. \[2010a\]](#) and [Hautsch and Huang \[2012\]](#), and references therein) and allows us to capture the trade-off between large-size and small-size limit orders. The temporary PIF is (as in [Almgren and Chriss \[2001\]](#) and [Kharroubi and Pham \[2010\]](#)) a function of order size and the time since the last trade. This simplest type of temporary price impact is far more tractable, both analytically and computationally, than that of [Obizhaeva and Wang \[2005\]](#) and [Gatheral et al. \[2011\]](#), which have a dependence on past order sizes.

Another important aspect we have to consider is the size and arrival rate of market orders since this determines the liquidity available for executing limit orders. Following [Avellaneda and Stoikov \[2008\]](#), we assume that market orders arrive according to a Poisson process. The size distribution of market orders can be very general, we only require that it is sufficiently smooth and decays sufficiently rapidly at infinity.

We formulate the optimal execution problem as an optimal stochastic control problem with three controls and three state variables. One control is the number of market sell orders submitted per time instant, and the two others are the size of the limit sell orders and the price at which they are posted relative to the mid-quote price. The three state variables are the time since the last trade, the shares remaining to sell and the time left for trading. As in [Almgren \[2009\]](#) and [Forsyth et al. \[2012\]](#) among others, our objective is to maximize the expected exponential utility. Dynamic programming is applied to derive a Hamilton-Jacobi-Bellman (HJB) equation. This first-order partial integro-differential equation (PIDE) is solved using the alternative directions implicit (ADI) finite-difference method and the (inverse) discrete fast Fourier transform (FFT).

We illustrate the model in a case study using order book data on two stocks. This case study indicates that limit orders have a significant impact on the optimal strategy regarding market orders. Instead of starting the execution with a relatively large amount of market orders, the optimal strategy is to wait for limit orders to be filled. Trading is subsequently complemented through market orders, depending on the rate at which limit orders are being filled. The price and size of limit orders depends on the risk aversion as well as the magnitude of the position and the urgency to execute it.

The paper is organized as follows. In [Section 1](#), we present the main building blocks of the model: the asset price dynamics and the liquidity available for executing market and limit orders. In [Section 2](#), we use dynamic programming to derive a PIDE that needs to be satisfied by the value function of the agent. In [Section 3](#), we discuss the numerical procedure to solve the PIDE. In [Section 4](#), we present a case study of our model. [Section 5](#) concludes the paper.

1 Optimal Execution

The optimal execution problem consists of liquidating large asset positions during a fixed period of time. In this study, we approach this problem from the standpoint of a risk-averse agent who can trade market and limit orders. The goal of the optimization is to resolve the following trade-off: If the agent liquidates the position in large chunks, there will be a noticeable price impact. This can be reduced by either splitting large market orders into a number of smaller ones, trading limit orders, or a combination of both. In the first case the agent will be exposed to price fluctuations, while in the second case the order may not be (entirely) executed due to a lack of liquidity. While previous research has predominantly focused on how to split market orders in an optimal manner, we add another dimension by analyzing the impact of limit orders.

The advantage of limit orders over market orders is that they are less affected by adverse price effects. Limit orders allow a pre-specified number of shares, the limit size, to be executed at a pre-specified price, the limit price, but do not guarantee an immediate or certain execution. Hence limit orders are useful when the price of execution is a priority over the certainty of execution, while the opposite holds true

for market orders. In the optimal execution problem, the certainty of execution is by its very definition a requirement, hence any execution strategy that is based solely on trading limit orders would not be feasible. However, we will show that the combined use of limit and market orders improves upon pure market order strategies. In what follows we will derive such strategies and discuss their properties.

In order to fully specify the optimal execution problem, we need a model for the risky asset and the liquidity available for executing market and limit orders. Furthermore, we have to define the class of admissible trading strategies and the objective function that captures the afore-mentioned trade-off.

Problem Formulation

We assume that the agent wants to liquidate an asset position of X shares over the time horizon $[0, T]$, $T < \infty$. In the examples and throughout the text, $X > 0$, but all results can be easily carried over to the case $X < 0$.¹ The remaining number of shares to sell at time t is the path $x(t)$, with $x(0) = X$ and $x(T) = 0$. We assume that the agent can trade market and limit sell orders, but we will not explicitly indicate the type of trade, i.e., buy or sell, if there is no danger of confusion. Market orders are submitted continuously on $[0, T]$, i.e., there is always sufficient liquidity for their execution. In contrast, the liquidity for executing limit orders is restricted and depends on the size and arrival time of market orders. As a consequence, limit orders will be filled at discrete (random) points in time. We assume that the agent can revise his orders at any given time (and without cost), but only one limit order can be active at a time, i.e., only by canceling his current limit order the agent can create a new one.

Finally, we introduce the lag variable $\theta(t)$ tracking the time since the last trade prior to or at time t . We set $\theta(t)$ equal to 0 at time $t = 0$, i.e., $\theta(0) = 0$.² Later on, it will be passed as an argument to the price impact function (PIF). Although θ is understood to be different for the buy and sell side of the market, we will not state this explicitly, as long as no confusion arises.

Model

We present the details of the model. To start with, we consider a probability space $(\Omega, \mathcal{F}, P)$ endowed with a filtration $\mathbb{F} = \{\mathcal{F}(t)\}_{t \in [0, T]}$ which represents the information structure available to the agent. We assume that $\mathcal{F}(0)$ is trivial and that $\mathcal{F}(T) = \mathcal{F}$. We also suppose that $\mathbb{F}$ satisfies the usual conditions of right-continuity and completeness (see, for example, [Karatzas and Shreve \[1991\]](#)). All components of the model will be defined on the filtered probability space $(\Omega, \mathcal{F}, \mathbb{F}, P)$. By $\mathbb{E}_t[\cdot]$ we denote the conditional expectation operator with respect to the filtration $\mathcal{F}(t)$.

Trading strategies

A trading strategy is a triplet (v, χ, ϱ) , where $v = \{v(t)\}_{t \in [0, T]}$ is a $\mathbb{F}$ -progressively measurable process of bounded variation, and $\chi = \{\chi(t)\}_{t \in [0, T]}$ and $\varrho = \{\varrho(t)\}_{t \in [0, T]}$ are $\mathbb{F}$ -predictable processes: $v(t)$ is the submission rate of market orders, $\chi(t)$ is the size of limit orders, and $\varrho(t)$ is the gap between the limit price and the mid-quote price $S(t-)$.³ We say that a trading strategy (v, χ, ϱ) on $[0, T]$ is admissible, and we denote $(v, \chi, \varrho) \in \mathcal{A}(0, T)$, if the following conditions hold true P -a.s.:

$$(i) \ v(t) \geq 0, \quad (ii) \ 0 \leq \chi(t) \leq x(t), \quad (iii) \ \varrho(t) \geq \alpha, \quad (1)$$

¹The case $X > 0$ implies that the agent has to sell shares, whereas $X < 0$ implies that he has to buy shares.

²This means we assume a trade at time $t = 0$. We stress that this assumption can be relaxed at the cost of additional notation.

³We denote the left-hand limit and right-hand limit of a function $f(\cdot)$ at $t \in [0, T]$ by $f(t-)$ and $f(t+)$, respectively.

for all $t \in [0, T]$, and where $\alpha \geq 0$ is half the size of the bid-ask spread. Conditions (i) and (ii) appear naturally given the target to sell shares up to an amount equal to $x(t)$. Condition (iii) is imposed because placing limit orders inside the bid-ask spread won't be optimal in our model.

A triple $(v, \chi, \varrho) \in \mathcal{A}(0, T)$ constitutes an admissible *execution* strategy, and we denote $(v, \chi, \varrho) \in \hat{\mathcal{A}}(0, T)$, if, in addition to condition (1), it reaches the target $x(T) = 0$ P -a.s..

Remark. Since limit order execution is uncertain the only way to achieve $x(T) = 0$ is by relying solely on market orders when the maturity time T nears.

Price of risky asset

The price of the risky asset S (which we take as the mid-quote price) evolves according to an exogenous arithmetic Brownian motion and an endogenous factor that is due to the permanent price impact of trades. The Brownian motion is assumed to be a noise term reflecting other (exogenous) influences on the price.

Large buy or sell orders, either market or limit, create order flow imbalances. Market participants, aware of these imbalances, should respond to them by adjusting their prices. This intuition is confirmed empirically in several studies. Cont et al. [2010a] show that over short time horizons there is a strong link between price changes and order flow imbalance. Hautsch and Huang [2011, 2012] find that limit orders do have significant long-term effects on prices and temporarily decrease bid-ask spreads.

These empirical findings provide the motivation for the permanent price impact function (PIF) assumed in our model. This function is a linear function of the cumulative order flow imbalance (COFI)⁴. We define COFI as the difference between all buy orders and all sell orders (including market orders, limit orders and cancellations) during a certain period of time. This means that given a trading strategy (v, χ, ϱ) the COFI over the period $[0, t]$ is equal to the number of executed orders in the period $[0, t)$ plus the number of outstanding limit orders at time t (all with a minus sign because they are sell orders), i.e.,

$$COFI(t) = -X + x(t-) - \chi(t) + \text{net order flow of other agents.} \quad (2)$$

where the net order flow of other agents will be discussed in the next subsection.

Assuming that the price S responds linearly to COFI, then, for all $t \in [0, T]$, $S(t)$ is given by⁵

$$S(t) = S(0) + \varphi COFI(t) + \sigma W(t),$$

with $\varphi > 0$ and where $W(t)$ is a standard $\mathcal{F}(t)$ -adapted Brownian motion. The linear relationship between COFI and price used in this model is both analytically tractable and empirically viable⁶. In this model, a buy (sell) order drives up (down) the price and this effect remains (i.e. has no resilience) as long as the order is not cancelled.

As is standard in the literature (see, e.g., Almgren and Chriss [2001] and Almgren [2003, 2009]), the exogenous part of $S(t)$ is modeled as a Bachelier model (arithmetic Brownian motion). Bachelier's model is found to be a good approximation of short-term asset price dynamics (see, e.g., Schachermayer and Teichmann [2008]), and yields analytical tractability, but may lead to negative prices even without price impact. However, in reality even large asset positions are typically executed within days or hours. We can

⁴Since we model the price level (and not price changes) we cumulate order flow imbalances.

⁵Empirical work of Cont et al. [2010a] and Hautsch and Huang [2012] shows that order flows deeper in the order book contribute substantially less to price changes than order flow at the best bid and ask. This can be incorporated into the model by letting φ depend on the limit price ϱ , but for ease of exposition and to keep the notation manageable we do not do so here.

⁶Cont et al. [2010a] reject the hypothesis of a non-linear relation between order flow imbalance and price changes.

therefore focus on the short-term dynamics for which the probability of negative values is negligible when using realistic parameters.

We point out that the first two terms in (2) yield a permanent price impact similar to the one in [Almgren and Chriss \[2001\]](#). As in [Almgren and Chriss \[2001\]](#) this part cannot be avoided; the agent has to move X shares through the market, and independent of how he schedules his orders, creates a permanent price impact. The last term in the COFI (the net order flow of other agents) cannot be controlled by the agent either. However, the third term (the size of limit orders) can be chosen to modulate the price impact of limit orders.

Order execution

We start by specifying how limit orders get filled against market orders of other agents. This necessitates a definition of price impact for discretely submitted market orders. Our agent submits market orders at a continuous rate, however. Hence we need to use a discrete-to-continuous-time limit to find the corresponding price impact of continuous-time market order strategies.

*Limit order execution*⁷. The liquidity for executing limit orders is determined by the size and the arrival time of counterpart market orders. These two quantities along with the distance from the limit to the market price determine the likelihood of how fast limit orders are filled. In order to proceed we thus need to describe the size and the frequency of market orders submitted by other agents.

The size of market orders is denoted by q (a negative value indicates a sell order, and a positive value denotes a buy order) and its probability density function (PDF) and cumulative distribution function (CDF) are denoted by f_Q and F_Q , respectively. We assume that $f_Q : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ decays sufficiently rapidly at infinity. This is to ensure that all upcoming integrals are well-defined. We furthermore assume that $F_Q(0) = \frac{1}{2}$ (so that buy and sell orders are evenly likely).

Following [Avellaneda and Stoikov \[2008\]](#) and [Cont et al. \[2010b\]](#), among others, we model the frequency at which market orders arrive as an independent $\mathcal{F}(t)$ -adapted Poisson process $N(t)$ with intensity λ .

Market order execution. The liquidity available for executing market orders is described by a temporary PIF that depends on the number of shares specified in the order e as well as the time θ since the last buy ($e > 0$) or sell order ($e < 0$) got executed. As in [Almgren and Chriss \[2001\]](#), we assume the PIF to be a function of these two quantities through $\xi := e/\theta$. This quotient can be interpreted as the rate of trading.

Denoting the temporary PIF by ψ the price at which a market order gets executed is given by

$$S(t) + \psi(\xi), \tag{3}$$

for all $t \in [0, T]$. We do not a priori assume a particular functional form on $\psi : \mathbb{R} \rightarrow \mathbb{R}$, but only require that ψ satisfies certain conditions, namely:

- (i) $\psi(0-) = -\alpha$, $\psi(0) = 0$, $\psi(0+) = \alpha$, and $\psi'(\xi) > 0$ on $\mathbb{R}/\{0\}$,
- (ii) $\Psi : \xi \mapsto \psi(\xi) + \xi\psi'(\xi)$ is C^1 on $\mathbb{R}/\{0\}$ and $\Psi'(\xi) > 0$ on $\mathbb{R}/\{0\}$,

where a prime denotes a first-order derivative. We stress that Ψ is the marginal cost of trading one more share, or put differently, the price at which the last share of a market order executes. Condition (i) reflects the bid-ask spread (of size 2α), and, in addition, says that a large-size market order has a higher (absolute) price impact than a small-size one; condition (ii) says that the marginal cost of price impact is increasing in order size.

⁷This part is inspired by the work of [Avellaneda and Stoikov \[2008\]](#).

In this model, the timing of a trade affects the price at which the next trade will be. In this sense, the price impact of a trade is temporary. In contrast to the models in [Obizhaeva and Wang \[2005\]](#), [Alfonsi et al. \[2010\]](#) and [Gatheral et al. \[2011\]](#), the size of a trade does not affect the price paid at the next transaction. In other words, prices are moderately resilient with respect to the timing of a trade and highly resilient when it comes to the size of a trade. This corresponds to an order book with instantaneous resilience and depth depending on the length of time between trades.

Modeling price impact as a function of the trading rate is quite common in the literature (see, e.g., [Almgren and Chriss \[2001\]](#), [Kharroubi and Pham \[2010\]](#) and [Forsyth \[2011\]](#), among others). Such models are more tractable than those of [Obizhaeva and Wang \[2005\]](#), [Alfonsi et al. \[2010\]](#) and [Gatheral et al. \[2011\]](#) and better reflect reality than those with PIFs that do not depend on the time interval between trades. However, as pointed out in, e.g., [Obizhaeva and Wang \[2005\]](#), they do not yield unique strategies that minimize the cost of executing certain amounts of shares. The common practice is to establish uniqueness by considering only continuous-time strategies. The fact that we consider other agents submitting market orders allows us to give a more proper justification. We show via [Proposition 1](#) that within the context of our model splitting up an order is rewarding.

Proposition 1. *Assume that there are other agents submitting market buy (sell) orders over any time interval $(s, t]$ ($0 \leq s < t \leq T$) with positive probability. Then the expected price impact cost of a market buy (sell) order can be reduced by breaking up the order into smaller pieces.*

The following proposition expresses the price impact of our continuous-time market sell order strategy $\{v(t)\}_{t \in [0, T]}$ in terms of ψ and is proven, together with [Proposition 1](#), in [Appendix A](#).

Proposition 2. *The price impact of an admissible continuous-time market sell order strategy $\{v(t)\}_{t \in [0, T]}$ at time $t \in [0, T]$ is given by*

$$\psi(-v(t)).$$

Since in the rest of the paper we deal only with continuous-time market sell order strategies, θ from now on refers to the time elapsed since the last market buy order arrived.

Implications of the model

Now that we have specified the model we can obtain an expression for $x(t)$ and $\theta(t)$ and see how they are controlled by the triplet (v, χ, ϱ) . First we note that an incoming market order of size $q(t)$ triggers limit orders with limit prices up to $\Psi(q(t)/\theta(t-))$ (which, we recall, is the price at which the last share of a market order of size $q(t)$ executes). This means that for a given trading strategy (v, χ, ϱ) , the number of matches between market and limit orders up to time $t \leq T$ can be written as

$$\int_0^t \mathbf{1}_{\{\Psi(q(s)/\theta(s-)) > \varrho(s)\}} dN(s). \quad (4)$$

As in [Avellaneda and Stoikov \[2008\]](#), [Guéant et al. \[2012\]](#) and [Guilbaud and Pham \[2013\]](#), the number of times limit orders are executed against incoming market orders turns out to be a Cox process with an intensity depending on the limit price. Besides, there is also a dependence on θ , which can be described as follows: when the last market order arrived not long ago (θ is small) limit orders are more likely to execute, and vice versa. This is a realistic feature, because recently depleted limit orders are not likely to be replaced immediately, which increases the chances of limit orders that are still standing to be filled.

When a limit order is triggered at time $t \in [0, T]$, $(q(t) - \theta(t-)\Psi^{-1}(\varrho(t))) \wedge \chi(t)$ shares are filled. The quantity $\theta(t-)\Psi^{-1}(\varrho(t))$ can be interpreted as the liquidity supplied by limit orders with more aggressive

limit prices. The number of shares filled with limit orders by time $t \in [0, T]$ is, hence, given by

$$\int_0^t ((q(s) - \theta(s-) \Psi^{-1}(\varrho(s)))_+ \wedge \chi(s)) dN(s),$$

where the “+” denotes the positive part and replaces the indicator function in (4).

The portfolio position at time $t \in [0, T]$ is given by

$$x(t) = x(s) - \int_s^t v(u) du - \int_s^t (q(u) - \theta(u-) \Psi^{-1}(\varrho(u)))_+ \wedge \chi(u) dN(u), \quad (5)$$

for all $s \in [0, t]$, where the first integral captures the amount of shares sold with market orders over the period $(s, t]$.

The value of the trade lag θ at time $t \in [0, T]$ is expressed as

$$\theta(t) = \theta(s) + t - s - \int_s^t \theta(u-) \mathbf{1}_{\{q(u) > 0\}} dN(u), \quad (6)$$

for all $s \in [0, t]$. We see that $\theta(t)$ moves with time but it resets back to zero when a market buy order is submitted.

Substituting the amount of market orders submitted by other market participants into (2) gives

$$COFI(t) = -X + x(t-) - \chi(t) + \int_0^t q(s) dN(s).$$

Hence the price $S(t)$ follows a jump-diffusion process.

Let $\mathcal{B}$ denote the cash holdings of the agent. Adding the cash proceeds of the sale of shares to the initial cash holdings $\mathcal{B}(0)$, we obtain

$$\mathcal{B}(t) = \mathcal{B}(0) + \int_0^t (S(s-) + \psi(-v(s))) v(s) ds + \int_0^t (S(s-) + \varrho(s)) ((q(s) - \theta(s-) \Psi^{-1}(\varrho(s)))_+ \wedge \chi(s)) dN(s), \quad (7)$$

where the left and right integral correspond to the amount of cash generated from the market and limit order execution. Applying integration by parts for càdlàg processes to (7) allows us to write the cash holdings after liquidation as

$$\begin{aligned} \mathcal{B}(T) &= \mathcal{B}(0) + XS(0) - \frac{1}{2} \varphi X^2 + \int_0^T \sigma x(t) dW(t) + \int_0^T \varphi x(t) q(t) dN(t) \\ &\quad + \int_0^T (\psi(-v(t)) - \varphi \chi(t)) v(t) dt + \int_0^T (\varrho(t) - \varphi \chi(t)) ((q(t) - \theta(t-) \Psi^{-1}(\varrho(t)))_+ \wedge \chi(t)) dN(t). \end{aligned}$$

The term $\mathcal{B}(0) + XS(0)$ stands for the market value of the portfolio at the beginning of the period.

Objective function

We assume that the agent is an expected exponential utility maximizer with a CARA (constant absolute risk aversion) coefficient $\gamma > 0$. Hence his objective function takes the form

$$\mathbb{E}_0[1 - \exp(-\gamma \mathcal{B}(T))]. \quad (8)$$

The exponential utility function brings several advantages of tractability and since, for the special case of $\lambda = 0$ (no liquidity to fill limit orders), maximizing (8) is equivalent to maximizing a mean-variance criterion⁸, we will be able to readily compare our results to papers that rely on the latter approach (see, for example, [Almgren and Chriss \[2001\]](#), [Almgren \[2003\]](#) and [Forsyth et al. \[2012\]](#)).

By substituting (7) into (8), we can rewrite the objective function as

$$\mathbb{E}_0 \left[1 - \exp \left(-\gamma \left(\mathcal{B}(0) + XS(0) - \frac{1}{2} \varphi X^2 + \int_0^T \sigma x(t) dW(t) + \int_0^T \varphi x(t) q(t) dN(t) + \int_0^T (\psi(-v(t)) - \varphi \chi(t)) v(t) dt + \int_0^T (\varrho(t) - \varphi \chi(t)) ((q(t) - \theta(t-) \Psi^{-1}(\varrho(t)))_+ \wedge \chi(t)) dN(t) \right) \right) \right]. \quad (9)$$

In Section 2, we will proceed by maximizing (9) over the control variables v , χ and ϱ using a dynamic programming approach. In the following we will neglect the constant 1 and the combination of terms $\mathcal{B}(0) + XS(0) - \frac{1}{2} \varphi X^2$ because all are irrelevant for the maximization problem (CARA utility allows us to factor out constants). Since $\mathcal{B}(0) + XS(0) - \frac{1}{2} \varphi X^2$ are the cash holdings after X is sold as a market order at $t = 0$ in the case of no liquidity constraint (i.e. $\psi(\xi) = 0$ for all $\xi \in \mathbb{R}$), the collection of terms that remain in the exponent of (9) stand for the cost of execution relative to this benchmark.

2 Hamilton-Jacobi-Bellman (HJB) Equation

In this section, we apply the dynamic programming technique to find the optimal control triplet (v, χ, ϱ) as a function of time t and space variables x and θ . We introduce the time-space domain $\Upsilon \times \Gamma \times \Theta$ in which (t, x, θ) evolves, with

$$\Upsilon = [0, T], \quad \Gamma = [0, X], \quad \text{and} \quad \Theta = [0, T].$$

The value function u is defined as the maximal attainable objective function, which can be easily expressed as the following minimization problem: for $x \in \Gamma$,

$$u(x) := \min_{(v, \chi, \varrho) \in \hat{\mathcal{A}}(0, T)} \mathbb{E}_0 \left[\exp \left(-\gamma \left(\int_0^T \sigma x(t) dW(t) + \int_0^T \varphi x(t) q(t) dN(t) + \int_0^T (\psi(-v(t)) - \varphi \chi(t)) v(t) dt + \int_0^T (\varrho(t) - \varphi \chi(t)) ((q(t) - \theta(t-) \Psi^{-1}(\varrho(t)))_+ \wedge \chi(t)) dN(t) \right) \right) \right],$$

where with a slight abuse of notation $x(t)$ and $\theta(t)$ are given by (5) and (6), respectively, with $s = 0$, $x(0) = x$ and $\theta(0) = 0$.

To invoke the dynamic programming principle, we introduce the time-dependent value function $\hat{u}$: for $(t, x, \theta) \in \Upsilon \times \Gamma \times \Theta$,

$$\hat{u}(t, x, \theta) := \min_{(v, \chi, \varrho) \in \hat{\mathcal{A}}(t, T)} \mathbb{E}_t \left[\exp \left(-\gamma \left(\int_t^T \sigma x(s) dW(s) + \int_t^T \varphi x(s) q(s) dN(s) + \int_t^T (\psi(-v(s)) - \varphi \chi(s)) v(s) ds + \int_t^T (\varrho(s) - \varphi \chi(s)) ((q(s) - \theta(s-) \Psi^{-1}(\varrho(s)))_+ \wedge \chi(s)) dN(s) \right) \right) \right], \quad (10)$$

⁸Since we deal with a non-Gaussian (transaction) risk (the number of limit orders that are executed over the period has a compound Cox distribution with a capped distribution of summands) and a non-Gaussian market risk ($S(t)$ follows a jump-diffusion), the classical “mean-variance” criterion is not an appropriate framework for analyzing optimal trading strategies that balance price impact and the range of risks involved.

where $x(t)$ and $\theta(t)$ are now given by taking $s = t$, $x(t) = x$ and $\theta(t) = \theta$ in (5) and (6), respectively.

In order to derive the Hamilton-Jacobi-Bellman (HJB) equation for the control problem formulated above, we rewrite (10) as follows

$$\begin{aligned} \widehat{u}(t, x, \theta) = \min_{(v, \chi, \varrho) \in \widehat{A}(t, T)} \mathbb{E}_t \left[\widehat{u} \left(\nu, x - \int_t^\nu v(s) ds - (q(\nu) - \theta(\nu-) \Psi^{-1}(\varrho(\nu)))_+ \wedge \chi(\nu), \theta(\nu) \right) \right. \\ \left. \exp \left(-\gamma \left(\int_t^\nu \sigma x(s) dW(s) + \varphi x(\nu) q(\nu) + \int_t^\nu (\psi(-v(s)) - \varphi \chi(s)) v(s) ds \right. \right. \right. \\ \left. \left. \left. + (\varrho(\nu) - \varphi \chi(\nu)) (q(\nu) - \theta(\nu-) \Psi^{-1}(\varrho(\nu)))_+ \wedge \chi(\nu) \right) \right) \right], \end{aligned} \quad (11)$$

where ν is the first time after time $t \in [0, T]$ that a market order is submitted, i.e., $\nu := \inf\{u \geq t : N(u) - N(t) = 1\} \wedge T$. By working out the expectation in (11), we obtain

$$\begin{aligned} \widehat{u}(t, x, \theta) = \min_{(v, \chi, \varrho) \in \widehat{A}(t, T)} \int_t^\infty \left[\int_{-\infty}^\infty \widehat{u} \left(\nu, x - \int_t^\nu v(s) ds - (q - \theta(\nu-) \Psi^{-1}(\varrho(\nu)))_+ \wedge \chi(\nu), \theta(\nu) \right) \right. \\ \left. \exp \left(\int_t^\nu \frac{\gamma^2 \sigma^2}{2} x^2(s) ds - \gamma \varphi x(\nu) q - \gamma \int_t^\nu (\psi(-v(s)) - \varphi \chi(s)) v(s) ds \right. \right. \\ \left. \left. - \gamma (\varrho(\nu) - \varphi \chi(\nu)) (q - \theta(\nu-) \Psi^{-1}(\varrho(\nu)))_+ \wedge \chi(\nu) \right) f_Q(q) dq \right] \lambda \exp(-\lambda(\nu - t)) d\nu. \end{aligned} \quad (12)$$

Define the first-order partial integro-differential operator $\mathcal{L}^{(v, \chi, \varrho)}$ by: for $(t, x, \theta) \in \Upsilon \times \Gamma \times \Theta$,

$$\begin{aligned} \mathcal{L}^{(v, \chi, \varrho)} \widehat{u} := \\ -v \widehat{u}_x + \frac{\widehat{u}_\theta}{2} + \left[\frac{\gamma^2 \sigma^2}{2} x^2 - \gamma (\psi(-v) - \varphi \chi) v \right] \widehat{u} - \lambda [(1 - F_Q^e(x, 0)) \widehat{u} - (F_Q^e(x, \infty) - F_Q^e(x, 0)) \widehat{u}_0] \\ + \lambda \int_0^\infty [\widehat{u}(t, x - q \wedge \chi, 0) \exp(-\gamma(\varrho - \varphi \chi)(q \wedge \chi)) - \widehat{u}_0] \exp(-\gamma \varphi x(q + \theta \Psi^{-1}(\varrho))) f_Q(q + \theta \Psi^{-1}(\varrho)) dq, \end{aligned} \quad (13)$$

where $\widehat{u}$ and $\widehat{u}_0$ is short-hand notation for $\widehat{u}(t, x, \theta)$ and $\widehat{u}(t, x, 0)$, and where subscripts denote partial derivatives. The function $F_Q^e : \Gamma \times [0, \infty] \rightarrow \mathbb{R}_{\geq 0}$ is an auxiliary function (to simplify notation) and is defined as

$$F_Q^e(x, y) = \int_{-\infty}^y e^{-\gamma \varphi x q} f_Q(q) dq.$$

In the rest of this work we suppress the dependence of $\mathcal{L}^{(v, \chi, \varrho)}$ on v , χ and ϱ and just write $\mathcal{L}$ for $\mathcal{L}^{(v, \chi, \varrho)}$. Define $A := [0, \infty] \times [0, x] \times [\alpha, \infty]$. Then following standard dynamic programming arguments, the solution to control problem (10) is given by the solution of the corresponding HJB equation⁹

$$0 = \widehat{u}_t + \min_{(v, \chi, \varrho) \in A} \mathcal{L} \widehat{u},$$

which, after inserting (13), yields the following first-order partial integro-differential equation (PIDE): for $(t, x, \theta) \in \Upsilon \times \Gamma \times \Theta$,

$$\begin{aligned} 0 = \widehat{u}_t \\ + \min_{(v, \chi, \varrho) \in A} \left\{ -v \widehat{u}_x + \frac{\widehat{u}_\theta}{2} + \left[\frac{\gamma^2 \sigma^2}{2} x^2 - \gamma (\psi(-v) - \varphi \chi) v \right] \widehat{u} - \lambda [(1 - F_Q^e(x, 0)) \widehat{u} - (F_Q^e(x, \infty) - F_Q^e(x, 0)) \widehat{u}_0] \right. \\ \left. + \lambda \int_0^\infty [\widehat{u}(t, x - q \wedge \chi, 0) \exp(-\gamma(\varrho - \varphi \chi)(q \wedge \chi)) - \widehat{u}_0] \exp(-\gamma \varphi x(q + \theta \Psi^{-1}(\varrho))) f_Q(q + \theta \Psi^{-1}(\varrho)) dq \right\}. \end{aligned}$$

⁹A justification is given in Appendix B.

By the change of variable $\tau := T - t$ for t , we convert the above forward equation into the following backward equation. Let $\tilde{u}(\tau, x, \theta) := \hat{u}(T - \tau, x, \theta)$. Then we obtain: for $(\tau, x, \theta) \in \Upsilon \times \Gamma \times \Theta$,

$$0 = -\tilde{u}_\tau + \min_{(v, \chi, \varrho) \in A} \left\{ -v\tilde{u}_x + \frac{\tilde{u}_\theta}{2} + \left[\frac{\gamma^2 \sigma^2}{2} x^2 - \gamma(\psi(-v) - \varphi\chi)v \right] \tilde{u} - \lambda \left[(1 - F_Q^e(x, 0))\tilde{u} - (F_Q^e(x, \infty) - F_Q^e(x, 0))\tilde{u}_0 \right] + \lambda \int_0^\infty [\tilde{u}(\tau, x - q \wedge \chi, 0) \exp(-\gamma(\varrho - \varphi\chi)(q \wedge \chi)) - \tilde{u}_0] \exp(-\gamma\varphi x(q + \theta\Psi^{-1}(\varrho))) f_Q(q + \theta\Psi^{-1}(\varrho)) dq \right\}. \quad (14)$$

Since this PIDE has no closed-form solution (to the best of our knowledge) we solve it using numerical methods in the next section.

Existence and uniqueness of a solution to the minimization problem in (14) are addressed in Propositions 3 and 4, respectively, which are proved in Appendix A. Existence is established for the general case, while uniqueness is established on a subspace of A under the assumption that ψ is linear and that q has a double-exponential (or Laplace) distribution. We leave for future work the more challenging issue of uniqueness of a minimizer in the general case.

Proposition 3. *The minimization problem in (14) admits a solution.*

Proposition 4. *A solution to the minimization problem in (14) with $\psi(\xi) = \alpha + \beta\xi$ and $f_Q(q) = \frac{\kappa}{2} \exp(-\kappa|q|)$ is unique on a subspace of A (which is defined in the proof).*

3 Numerical Solution

We apply the finite-difference (FD) alternating-directions implicit (ADI) method (hereafter FD-ADI method) to solve the PIDE (14). The FD-ADI method is an efficient way of reducing higher dimensional problems to a sequence of one-dimensional problems. The FD-ADI method was first introduced by Peaceman and Rachford [1955] and Douglas and Rachford [1956], as a perturbation of the Crank-Nicolson scheme. The main idea of the FD-ADI method is to split the operator $\mathcal{L}$ into n sub-operators and subsequently to apply the Crank-Nicolson scheme n times, at each time treating only one sub-operator implicitly (for more details about the ADI method we refer to Duffy [2006]). The FD-ADI method allows us to improve the computational efficiency, but does not address another potential (numerical) issue. The discretization of the integral term in (14) leads to a non-sparse matrix that has to be inverted in the ADI scheme. We avoid this costly task by relying on a (inverse) fast Fourier transform (FFT) method.

We would like to point out that to analytically determine the optimal control (by differentiating (14) and setting the result to zero) and then to discretize the resulting PIDE leads to a PIDE that is highly non-linear for which it is extremely difficult to create a converging numerical scheme. Therefore, we discretize PIDE (14) first, and then optimize the resulting equations over v , χ and ϱ . For more information about this issue and numerical methods for HJB equations in general, we refer to Forsyth [2011].

We start by splitting the operator $\mathcal{L}$ into five parts. First, $\mathcal{L}$ is split into $\mathcal{H}$, $\mathcal{D}$ and $\mathcal{I}$ and we write $\mathcal{L} = \mathcal{H} + \mathcal{D} + \mathcal{I}$. Here $\mathcal{H}$ captures the non-differential part in (14), i.e.,

$$\mathcal{H}\tilde{u} = \left[\frac{\gamma^2 \sigma^2}{2} x^2 - \gamma(\psi(-v) - \varphi\chi)v \right] \tilde{u},$$

and $\mathcal{D}$ and $\mathcal{I}$ stand for the diffusion and integral parts of $\mathcal{L}$. Second, each sub-operator $\mathcal{D}$ and $\mathcal{I}$ is split into two parts: $\mathcal{D} = \mathcal{D}_1 + \mathcal{D}_2$ and $\mathcal{I} = \mathcal{I}_1 + \mathcal{I}_2$, where $\mathcal{D}_1$ and $\mathcal{I}_1$ ($\mathcal{D}_2$ and $\mathcal{I}_2$) are operators in the x -direction

(θ -direction). The sub-operators $\mathcal{D}_1$, $\mathcal{D}_2$, $\mathcal{I}_1$ and $\mathcal{I}_2$ can be written as follows:

$$\begin{aligned}\mathcal{D}_1 \tilde{u} &= -v \frac{\partial \tilde{u}}{\partial x}, & \mathcal{D}_2 \tilde{u} &= \frac{1}{2} \frac{\partial \tilde{u}}{\partial \theta}, & \mathcal{I}_2 \tilde{u} &= -\lambda [(1 - F_Q^e(x, 0))\tilde{u} - (F_Q^e(x, \infty) - F_Q^e(x, 0))\tilde{u}_0], \\ \mathcal{I}_1 \tilde{u} &= \lambda \int_0^\infty [\tilde{u}(\tau, x - q \wedge \chi, 0) \exp(-\gamma(\varrho - \varphi\chi)(q \wedge \chi)) - \tilde{u}_0] \exp(-\gamma\varphi x(q + \theta\Psi^{-1}(\varrho))) f_Q(q + \theta\Psi^{-1}(\varrho)) dq.\end{aligned}$$

Having split the operator $\mathcal{L}$, we now consider the problem of finding $\tilde{u} : \Upsilon \times \Gamma \times \Theta \rightarrow \mathbb{R}$ such that

$$\frac{\partial \tilde{u}}{\partial \tau} = \min_{(v, \chi, \varrho) \in A} \{ \mathcal{H}\tilde{u} + \mathcal{D}_1 \tilde{u} + \mathcal{D}_2 \tilde{u} + \mathcal{I}_1 \tilde{u} + \mathcal{I}_2 \tilde{u} \}, \quad \text{in } \Upsilon \times \Gamma \times \Theta, \quad (15)$$

$$\tilde{u}(\tau, 0, \theta) = 1, \quad \text{in } \Upsilon \times \Theta, \quad (16)$$

$$\frac{\partial \tilde{u}}{\partial \theta}(\tau, x, T) = 0, \quad \text{in } \Upsilon \times \Gamma, \quad (17)$$

where the boundary condition (16) is of Dirichlet type and reflects the fact that no further cost is incurred after the portfolio is liquidated; and where (17) is a Neumann boundary condition stating that $\tilde{u}$ becomes insensitive to θ for large values of θ . The initial condition is set equal to the leading term of (10) with $v = x/\tau$ in the limit of $\tau \rightarrow 0$, i.e., for $(x, \theta) \in \Gamma \times \Theta$

$$\tilde{u}(\tau, x, \theta) \sim \exp\left(\frac{\gamma\psi(x)x}{\tau}\right) \quad \text{for } \tau \rightarrow 0. \quad (18)$$

Since this asymptotic expression gets arbitrarily large for positive x , the agent is forced to reach his execution target.

We discretize the domain¹⁰ $\Upsilon \times \Gamma \times \tilde{\Theta}$, where $\tilde{\Theta} := [0, \theta_{\max}]$ and $\theta_{\max} \leq T$, using the equally-spaced grid points

$$\begin{aligned}\tau_m &:= mk, & m &= 1, 2, \dots, M, & k &:= T/M = \Delta\tau, \\ x_i &:= ih_1, & i &= 0, 1, \dots, N_1, & h_1 &:= X/N_1 = \Delta x, \\ \theta_j &:= jh_2, & j &= 0, 1, \dots, N_2, & h_2 &:= \theta_{\max}/N_2 = \Delta\theta.\end{aligned}$$

Let $\tilde{u}_{i,j}^m$ denote the discrete approximation to the exact solution $\tilde{u}(\tau_m, x_i, \theta_j)$ and let $(v_{i,j}^m, \chi_{i,j}^m, \varrho_{i,j}^m)$ denote the approximate value of the control triplet (v, χ, ϱ) at the grid point (τ_m, x_i, θ_j) . The quantities $\tilde{u}_{i,j}^m$, $v_{i,j}^m$, $\chi_{i,j}^m$ and $\varrho_{i,j}^m$ are stacked together to form the $(N_1 + 1) \times (N_2 + 1)$ matrices $\underline{\tilde{u}}^m := \{\tilde{u}_{i,j}^m\}_{j=0, \dots, N_2}^{i=0, \dots, N_1}$, $\underline{v}^m := \{v_{i,j}^m\}_{j=0, \dots, N_2}^{i=0, \dots, N_1}$, $\underline{\chi}^m := \{\chi_{i,j}^m\}_{j=0, \dots, N_2}^{i=0, \dots, N_1}$ and $\underline{\varrho}^m := \{\varrho_{i,j}^m\}_{j=0, \dots, N_2}^{i=0, \dots, N_1}$. The operators $\mathcal{D}_1$ and $\mathcal{D}_2$ are replaced with finite-difference approximations D_1 and D_2 , which are bi-diagonal matrices of size $(N_1 + 1) \times (N_1 + 1)$ and $(N_2 + 1) \times (N_2 + 1)$. For the τ - and x -derivative, we apply forward differencing, while backward differencing is applied to the θ -derivative. This is to ensure that our numerical scheme has positive coefficients. This is not guaranteed when using central differencing, which would otherwise be more desirable because it implies second-order (instead of first-order) accuracy. A detailed discussion of this issue can be found in Wang and Forsyth [2008] or Forsyth and Vetzal [2011]. The integral operator $\mathcal{I}_1$, which depends on θ , is approximated by a trapezoidal quadrature rule to obtain the $(N_1 + 1) \times (N_1 + 1)$ matrices I_1^j ($j = 0, \dots, N_2$) where the superscript j refers to the value of θ ; and $\mathcal{I}_2$, which is a (simple) difference operator, is replaced by I_2 which maps $\tilde{u}_{i,j}^m$ to $-\lambda[(1 - F_Q^e(x_i, 0))\tilde{u}_{i,j}^m - (F_Q^e(x_i, \infty) - F_Q^e(x_i, 0))\tilde{u}_{i,1}^m]$. Finally, we introduce the $(N_1 + 1) \times (N_2 + 1)$ matrix H , which is the discretized version of the operator $\mathcal{H}$.

For time-stepping we use the ADI method. First, a half-step is taken implicitly in the x -direction and explicitly in the θ -direction followed by a half-step taken implicitly in the θ -direction and explicitly in the

¹⁰The original domain is truncated in the direction of θ in order to reduce the computational effort.

x -direction. When applied to problem (15)-(17), this yields the following equations:^{11,12,13}

$$\underline{u}^{m+\frac{1}{2}} = \underline{u}^m + \frac{\Delta\tau}{2} \left(\underline{u}^m (D_2 + I_2) + D_1 \underline{u}^{m+\frac{1}{2}} + [I_1^1 \underline{u}^{m+\frac{1}{2}} \dots I_1^{N_2} \underline{u}^{m+\frac{1}{2}}] + H \circ \underline{u}^m \right), \quad (19)$$

$$(\underline{v}^*)^{m+\frac{1}{2}} = \min_{\underline{v}^{m+\frac{1}{2}} \in V^{m+\frac{1}{2}}} \left\{ D_1 \underline{u}^{m+\frac{1}{2}} + H \circ \underline{u}^m \right\}, \quad (\chi^*)^{m+\frac{1}{2}} = (\chi^*)^m, \quad (\varrho^*)^{m+\frac{1}{2}} = \min_{\underline{\varrho}^{m+\frac{1}{2}} \in \Xi^{m+\frac{1}{2}}} \left\{ [I_1^1 \underline{u}^{m+\frac{1}{2}} \dots I_1^{N_2} \underline{u}^{m+\frac{1}{2}}] \right\},$$

$$\underline{u}^{m+1} = \underline{u}^{m+\frac{1}{2}} + \frac{\Delta\tau}{2} \left(\underline{u}^{m+1} D_2 + \underline{u}^{m+\frac{1}{2}} I_2 + D_1 \underline{u}^{m+\frac{1}{2}} + [I_1^1 \underline{u}^{m+\frac{1}{2}} \dots I_1^{N_2} \underline{u}^{m+\frac{1}{2}}] + H \circ \underline{u}^{m+\frac{1}{2}} \right), \quad (20)$$

$$(\underline{v}^*)^{m+1} = (\underline{v}^*)^{m+\frac{1}{2}}, \quad (\chi^*)^{m+1} = \min_{\underline{\chi}^{m+1} \in \Xi^{m+1}} \left\{ [I_1^1 \underline{u}^{m+\frac{1}{2}} \dots I_1^{N_2} \underline{u}^{m+\frac{1}{2}}] + H \circ \underline{u}^{m+\frac{1}{2}} \right\}, \quad (\varrho^*)^{m+1} = (\varrho^*)^{m+\frac{1}{2}},$$

where “ $\circ$ ” is the usual Hadamard product, and where $\underline{u}^{m+\frac{1}{2}}$ denotes the j -th column of $\underline{u}^{m+\frac{1}{2}}$. Furthermore, $\underline{v}^{m+\frac{1}{2}} \in V^{m+\frac{1}{2}}$, $\underline{\varrho}^{m+\frac{1}{2}} \in \Pi^{m+\frac{1}{2}}$ and $\underline{\chi}^{m+1} \in \Xi^{m+1}$ imply that $v_{i,j}^{m+\frac{1}{2}} \in V_{i,j}^{m+\frac{1}{2}} := [0, x_i/k]$, $\varrho_{i,j}^{m+\frac{1}{2}} \in \Pi_{i,j}^{m+\frac{1}{2}} := [\varrho_{\min}, \varrho_{\max}]$ and $\chi_{i,j}^{m+1} \in \Xi_{i,j}^{m+1} := [0, x_i]$ for all $i = 0, \dots, N_1$ and $j = 0, \dots, N_2$. The definition of $V_{i,j}^{m+\frac{1}{2}}$ and $\Xi_{i,j}^{m+1}$ ensures that the discretized state variable x_i stays inside the computational domain. As initial data we take the expression (18) at an initial time $\tau = \tau_1$, i.e., $\tilde{u}_{i,j}^1 = \exp(\gamma\psi(x_i)x_i/\tau_1)$ for all $i = 0, \dots, N_1$ and $j = 0, \dots, N_2$. The starting values for the controls are $(v_{i,j}^*)^1 = x_i/\tau_1$ and $(\chi_{i,j}^*)^1 = (\varrho_{i,j}^*)^1 = 0$ for all $i = 0, \dots, N_1$ and $j = 0, \dots, N_2$.

An explicit expression for the optimal control $(\underline{v}^{m+\frac{1}{2}})^*$ is obtained by differentiating $D_1 \underline{u}^{m+\frac{1}{2}} + H \circ \underline{u}^m$ with respect to $\underline{v}^{m+\frac{1}{2}}$ and setting the result to zero. Since the two other optimal controls cannot be obtained explicitly, they are found by linear search after discretizing the range of controls $\Pi^{m+\frac{1}{2}}$ and Ξ^{m+1} both into L elements (which are concentrated around the optimal values of the previous time step). This will introduce a small error whose significance disappears for large L due to the continuity of the minimization functions.

Equations (19)-(20) are equivalent to

$$\begin{aligned} \left(E_{N_1+1} - \frac{\Delta\tau}{2} D_1 \right) \underline{u}^{m+\frac{1}{2}} - \frac{\Delta\tau}{2} [I_1^1 \underline{u}^{m+\frac{1}{2}} \dots I_1^{N_2} \underline{u}^{m+\frac{1}{2}}] &= \underline{u}^m \left(E_{N_2+1} + \frac{\Delta\tau}{2} (D_2 + I_2) \right) + \frac{\Delta\tau}{2} (H \circ \underline{u}^m), \quad (21) \\ \underline{u}^{m+1} \left(E_{N_2+1} - \frac{\Delta\tau}{2} D_2 \right) &= \underline{u}^{m+\frac{1}{2}} \frac{\Delta\tau}{2} I_2 + \left(E_{N_1+1} + \frac{\Delta\tau}{2} D_1 \right) \underline{u}^{m+\frac{1}{2}} \\ &\quad + \frac{\Delta\tau}{2} [I_1^1 \underline{u}^{m+\frac{1}{2}} \dots I_1^{N_2} \underline{u}^{m+\frac{1}{2}}] + \frac{\Delta\tau}{2} (H \circ \underline{u}^{m+\frac{1}{2}}), \end{aligned}$$

where E_n denotes the n -dimensional identity matrix.

The matrices D_1 , D_2 and I_2 are sparse and can hence be inverted at a very low computational cost, as opposed to the non-sparse matrix I_1 . The former ensures the fast computation of the second to the last column of $\underline{u}^{m+\frac{1}{2}}$ and of $\underline{u}^{m+1}$, but the latter poses a serious numerical problem when computing $\underline{u}_{:,1}^{m+\frac{1}{2}}$ and N_1 is large. In case N_1 is large, the inversion of I_1 is therefore circumvented using the following inverse FFT technique. We rewrite the first column on the left-hand side (LHS) of (21) using the discrete version

¹¹Since matrix notation is used we write operators in the x -direction in front of the matrix $\underline{u}$, while operators in the θ -direction are written behind the matrix $\underline{u}$.

¹²We split the optimization problem into a minimization over v and ϱ (first half time step) and a minimization over χ (second half time step). This brings down the computational cost of the search for an optimizer dramatically. We verified that this splitting of the optimization problem does not influence our results in any way.

¹³Optimal values are denoted by a “ $*$ ”.

of the relation¹⁴

$$\int_{-\infty}^{\infty} \tilde{u}(x-q)\tilde{p}(q)dq = - \int_{-\infty}^{\infty} \tilde{u}(z)\tilde{p}(x-z)dz = -(\tilde{u} * \tilde{p})(x),$$

where $\tilde{u} : x \mapsto \tilde{u}(\tau, x, 0)$ and $\tilde{p} : q \mapsto p(x, 0, q)$, and where $p : \Gamma \times \Theta \times [0, \infty] \rightarrow \mathbb{R}$ is defined in such a way that the LHS above is equal to the integral in (14), i.e.,

$$p(x, \theta, q) := \lambda \exp(-\gamma(\varrho - \varphi\chi)q - \gamma\varphi x(q + \theta\Psi^{-1}(\varrho)))f_Q(q + \theta\Psi^{-1}(\varrho))\mathbf{1}_{\{0 \leq q \leq \chi\}} \\ + \delta(q - \chi)(F_Q^e(x, \infty) - F_Q^e(x, \chi + \theta\Psi^{-1}(\varrho))) - \delta(q)(F_Q^e(x, \infty) - F_Q^e(x, \theta\Psi^{-1}(\varrho))).$$

We obtain

$$\left(E_{N_1+1} - \frac{\Delta\tau}{2}D_1\right)\underline{\tilde{u}}_{\cdot,1}^{m+\frac{1}{2}} - \frac{\Delta\tau}{2}I_1^1\underline{\tilde{u}}_{\cdot,1}^{m+\frac{1}{2}} = \left(E_{N_1+1} - \frac{\Delta\tau}{2}D_1\right)\underline{\tilde{u}}_{\cdot,1}^{m+\frac{1}{2}} + \frac{\Delta\tau}{2}\left(\underline{\tilde{u}}_{\cdot,1}^{m+\frac{1}{2}} * \bar{I}_1\right), \quad (22)$$

where, for all $i = 0, \dots, N_1$ and $j = 0, \dots, N_2$, $(I_1^j)_{i+1,\cdot}$, respectively $(\bar{I}_1)_{\cdot,i+1}$, are $N_1 + 1$ point trapezoidal quadrature approximation of the functions $z \mapsto p(x_i, \theta_j, x_i - z)$ and $z \mapsto p(x_i, 0, z)$. The convolution in (22) is understood as a $(N_1 + 1)$ -dimensional vector whose i -th element is equal to $(\underline{\tilde{u}}_{\cdot,1}^{m+\frac{1}{2}} * (\bar{I}_1)_{\cdot,i})_i = \sum_{\tilde{i}=1}^i \tilde{u}_{i-\tilde{i},1}^{m+\frac{1}{2}}(\bar{I}_1)_{\tilde{i},i}$.

For computational convenience we restrict v to be equal to zero when $\theta = 0$,¹⁵ and hence the term with D_1 drops out. Of course, this introduces a small numerical error in the results, but we verified that for reasonably small mesh sizes this error can be neglected.

Denoting the first column on the RHS of (21) by y and taking the discrete Fourier transform (DFT) of the resulting equation we obtain

$$\text{DFT} \left[E_{N_1+1}\underline{\tilde{u}}_{\cdot,1}^{m+\frac{1}{2}} + \frac{\Delta\tau}{2}\left(\underline{\tilde{u}}_{\cdot,1}^{m+\frac{1}{2}} * \bar{I}_1\right) \right] = \text{DFT}[y]. \quad (23)$$

The convolution integral theorem allows us to rewrite the LHS of (23) as

$$\text{DFT} \left[\underline{\tilde{u}}_{\cdot,1}^{m+\frac{1}{2}} \right] + \frac{\Delta\tau}{2}\text{DFT} \left[\underline{\tilde{u}}_{\cdot,1}^{m+\frac{1}{2}} \right] \circ \text{diag}(\text{DFT} [\bar{I}_1]).$$

We obtain the following explicit expression for $\text{DFT}[\underline{\tilde{u}}_{\cdot,1}^{n+\frac{1}{2}}]$:

$$\text{DFT} \left[\underline{\tilde{u}}_{\cdot,1}^{m+\frac{1}{2}} \right] = \text{DFT}[y] \oslash \left(1 + \frac{\Delta\tau}{2}\text{diag}(\text{DFT}[\bar{I}_1]) \right),$$

where “ $\oslash$ ” denotes the Hadamard division (component-wise division). We can now take the inverse discrete Fourier transform to obtain $\underline{\tilde{u}}_{\cdot,1}^{m+\frac{1}{2}}$, i.e.,

$$\underline{\tilde{u}}_{\cdot,1}^{m+\frac{1}{2}} = \text{DFT}^{-1} \left[\text{DFT}[y] \oslash \left(1 + \frac{\Delta\tau}{2}\text{diag}(\text{DFT}[\bar{I}_1]) \right) \right]. \quad (24)$$

The computational cost associated with $\text{DFT}[\bar{I}_1]$ is of $O(N_1^2 \log_2 N_1)$ but it only needs to be computed once. The computational cost of solving (21) for $\underline{\tilde{u}}_{\cdot,1}^{m+\frac{1}{2}}$ using (24) given $\text{DFT}[\bar{I}_1]$, is one Hadamard

¹⁴This part is inspired by the work of Andersen and Andreasen [2000].

¹⁵In economic terms it means that we don't allow a market order to be submitted at the first time step after a limit order is hit.

division (of $O(N_1)$), one fast Fourier transform (FFT) and one inverse FFT, i.e., $O(N_1 \log_2 N_1)$, and hence, the total cost for obtaining $\tilde{u}_{\cdot,1}^{m+\frac{1}{2}}$ for all m is of $O(\max\{M, N_1\} N_1 \log_2 N_1)$,¹⁶ whereas with matrix inversion it is of $O(\max\{M, N_1\} N_1^2)$. The Fourier representation (24) ensures that the computation of $\tilde{u}_{\cdot,1}^{m+\frac{1}{2}}$ doesn't dominate the computational burden when N_1 gets large. However, we experienced that the Fourier transform method is moderately slower for small N_1 , and is found to be faster for $N_1 > 2^6$.

4 Numerical Illustrations

This section aims to study the properties of the value function $\tilde{u}$ and the optimal controls v^* , χ^* and ϱ^* . For this purpose, we compute some numerical examples using realistic parameters.

First we choose a particular PIF ψ and CDF F_Q of the size of incoming market orders. We assume that ψ is linear on $\mathbb{R}/\{0\}$, i.e.,

$$\psi(\xi) = \begin{cases} -\alpha + \beta\xi & \xi < 0 \\ 0 & \xi = 0 \\ \alpha + \beta\xi & \xi > 0 \end{cases}, \quad (25)$$

with the slope coefficient $\beta > 0$ being interpreted as a measure of illiquidity: the higher β , the higher the liquidity premium. Recall that α stands for half the size of the bid-ask spread. With this choice of ψ , we obtain $\Psi(\xi) = \psi(\xi) + \beta\xi$.

We assume that the size of market orders q has a double-exponential (or Laplace) distribution, i.e.,

$$f_Q(q) := \frac{\kappa}{2} \exp(-\kappa|q|) \quad \text{and} \quad F_Q(q) := \begin{cases} \frac{1}{2} \exp(\kappa q) & q < 0 \\ 1 - \frac{1}{2} \exp(-\kappa q) & q \geq 0 \end{cases}.$$

The choices for ψ and F_Q are relatively simple so that only a few parameters need to be estimated. This facilitates the analysis and the interpretation of results.

Now we need to determine values for 7 parameters, namely, σ , λ , φ , α , β , κ and γ . To set realistic values, we use limit order book data¹⁷ for Groupe Danone (GD) and Électricité de France (EDF), both listed on the Paris Bourse and member of the CAC 40 stock index. Table 1 contains descriptive statistics of this data set. All values (except the ones on the top two lines) are averages for March 2011.

¹⁶Note that this must be multiplied by L when optimizing χ and ϱ as described earlier.

¹⁷The order book data is provided by the BNP Paribas Chair of Quantitative Finance, École Centrale Paris, France.

	EDF	GD
price on 2011-03-01 (open) (in €)	32.35	45.67
price on 2011-03-31 (close) (in €)	29.22	46.10
# market orders (per hour)	1.31×10^3	1.40×10^3
size of market orders ($\times 100$ shrs.)	2.54	2.09
tick size (in € cents)	0.50	0.50
bid-ask spread (in € cents)	1.35	1.57
depth ask side ($\times 100$ shrs.)	65.02	40.52
depth bid side ($\times 100$ shrs.)	70.33	41.17
price diff. ask (in € cents)	2.40	2.70
price diff. bid (in € cents)	2.34	2.46

Table 1: Summary statistics of the limit order book data set. Depth refers to the pending volume at the best five price levels of the order book. The last two statistics give the price difference between the best quote and the fifth best quote at the ask side and the bid side, respectively.

We now turn to a concrete problem. Suppose we need to liquidate 2×10^4 EDF shares within 1 hour. Note that this is approximately 6% of the (average) hourly volume of $254 \times 1.31 \times 10^3 = 3.30 \times 10^5$ shares, which is a typical percentage for large-order executions. Measuring X in number of shares and time in hours we set $X = 2 \times 10^4$ and $T = 1$. EDF stock has an absolute volatility of $\sigma = 0.22$ (measured in euros per share per square-root hours). The arrival rate of market orders is assumed to be equal to the average number of market orders per hour. The parameter φ is obtained by a linear regression of price changes on order flow imbalances (measured by the difference between market depth on the ask side and the bid side)¹⁸. We obtain a number of 3.94×10^{-8} . The value for α is set equal to 6.75×10^{-3} , which is half the size of the bid-ask spread. The parameter β is less straightforward to determine. First we note that (25) implies a depth of $1/(2\zeta\beta)$ shares per Euro, when ζ market orders are submitted once every hour. Since the order book of EDF has an (average) depth of 3.01×10^5 (7033/0.0234) shares per Euro on the bid side¹⁹, and since market orders arrive at a rate of 1.31×10^3 per hour, we approximate β by $1/(2 \times ((3.01 \times 10^5) \times 1.31 \times 10^3/2)) = 2.53 \times 10^{-9}$; we divided 1.31×10^3 by 2 since we are only looking at the arrival of market sell orders. The parameter κ is obtained as one over the average size of market orders so that the expectation of q (conditional on $q > 0$) matches its empirical average. Finally we need to choose a value for the risk-aversion coefficient γ . Following Almgren and Chriss [2001], we choose γ such that the (expected) portfolio position after half the time has passed is (roughly) equal to 40% of its initial size. We present a summary of the 7 parameters for both stocks in Table 2, where the parameters for GD stock have been calculated in a similar way as has been done for EDF stock.

¹⁸See Cont et al. [2010a] for a fine empirical analysis of the relation between order flow imbalance and price changes.

¹⁹We use data on the bid side to determine β since we are submitting market sell orders.

	EDF	GD
X (in shrs.)	2×10^4	2×10^4
T (in hours)	1	1
σ (in (€/shrs.)/hour $^{\frac{1}{2}}$)	0.22	0.23
λ (in (# market orders/hour))	1.31×10^3	1.39×10^3
φ (in (€/shrs.))	3.94×10^{-8}	1.06×10^{-7}
α (in €)	6.75×10^{-3}	7.85×10^{-3}
β (in (€/shrs.)/(# market orders/hour))	2.53×10^{-9}	4.25×10^{-9}
κ	3.9×10^{-3}	4.8×10^{-3}
γ	2×10^{-7}	2×10^{-7}

Table 2: Summary of model parameter values for EDF stock and GD stock.

Since, in practice, an order book is a list of limit orders that are pending at certain price levels (which are at least one tick apart) and usually executed in time priority at a given price level²⁰, we subtract the depth at one tick from the size of an incoming market order q . In this way, we capture the fact that limit orders not only queue behind other limit orders at better price levels (price-priority), but also behind those at the same price level (time-priority) – which is not captured in the PIF specification (25). Put differently, it means that a limit sell order (of our agent) can only be triggered when the size of an incoming market buy order exceeds the depth at the best ask price. In mathematical terms, it means that “ $(q \wedge \chi)$ ” in (14) is replaced by $(q - \frac{\theta}{2\beta} \times 5 \times 10^{-3}) \wedge \chi$, where $\frac{\theta}{2\beta} \times 5 \times 10^{-3}$ is the depth at one tick in our model. We stress that this is a conservative way of accounting for the queuing delay since a limit order may also gain time priority.

Now we numerically solve (14) using the FD-ADI method as described in Section 3. The values for the numerical parameters are: $N_1 = N_2 = 2^7$, $L = 2^7$, $\theta_{\max} = 10^{-2}$ (hours), $\varrho_{\min} = 6.75 \times 10^{-3}$ (€) and $\varrho_{\max} = 2.5 \times 10^{-2}$ (€). We first perform a convergence analysis by computing the value function at a fixed point in space and time for four levels of mesh refinement. Table 3 lists the number of nodes and time-steps we use. The last column shows the corresponding entries of the value function. We observe first-order convergence of the numerical scheme in time and space, which is to be expected from a FD-ADI scheme based on forward and backward differences.

$\log_2(M)$	$\log_2(N_1)$	$\log_2(N_2)$	$\tilde{u}_{N_1/2, N_2/2}^{M/2} - 1$	$\Delta \tilde{u}_{N_1/2, N_2/2}^{M/2}$
5	5	5	1.1478×10^{-5}	/
6	6	6	1.2605×10^{-5}	1.1274×10^{-6}
7	7	7	1.3282×10^{-5}	0.6767×10^{-6}
8	8	8	1.3622×10^{-5}	0.3403×10^{-6}

Table 3: Entries of the numerical value function $\tilde{u}_{N_1/2, N_2/2}^{M/2}$ (which approximates $\tilde{u}(X/2, \theta_{\max}/2, T/2)$) for different levels of mesh refinement.

Figure 1 shows how the position evolves over time when the optimal execution strategy is followed. The order execution is illustrated with three lines. The thin gray line displays the solution to the execution problem where the agent can solely trade market orders. In this simplest case the optimal strategy is to start trading relatively fast and then gradually reducing the trading intensity over the course of the execution, as can be observed in Figure 1.

Once we account for limit orders the position at time $\tau \in (0, 1)$ ceases to be fully deterministic. Figure

²⁰This is, among others, the case for shares of EDF and GD traded on the Paris stock exchange.

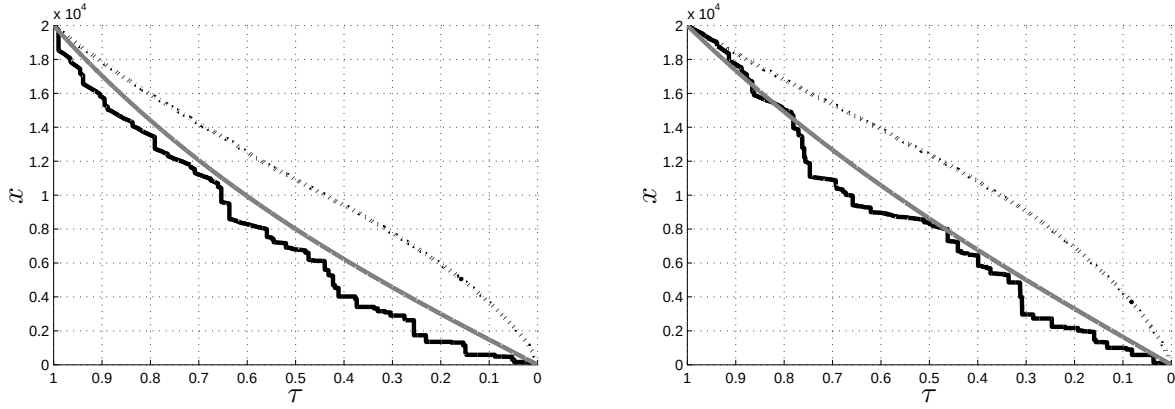


Figure 1: Evolution of the position in EDF stock (left graph) and GD stock (right graph).

1 highlights the impact of a specific realization of the market order arrival process $N(t)$ and the order sizes $q(t)$. The dotted line corresponds to a scenario where no limit orders are being executed, while the bold line shows the impact of limit orders that are being executed, visible as vertical drops.

The effect of limit orders on the optimal strategy is quite substantial. Since trading limit orders enables agents to potentially avoid adverse price effects they have an incentive to wait for them to be executed before increasingly relying on market orders²¹. This is illustrated in Figure 1 where the dotted line always lies above the trajectory that only assumes the possibility of trading market orders. The magnitude of this effect becomes more pronounced when any of the following occurs: (i) the liquidity for executing limit orders grows (i.e. a higher value for λ and/or a lower value for κ), (ii) the cost of trading market orders increases relative to the cost of trading limit orders (i.e. a higher value for β and/or a lower value for φ), or (iii) the bid-ask spread widens (a higher value for α). A lower value for α and β largely explains the weaker effect of limit orders on the optimal strategy for selling EDF stock as compared to GD stock.

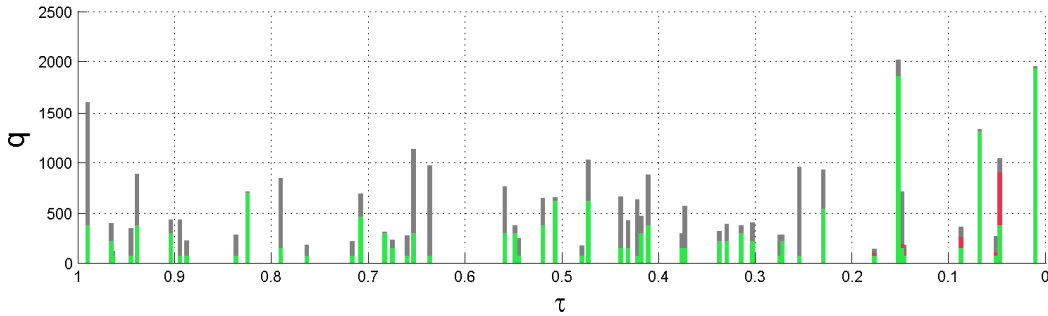


Figure 2: The size and arrival time of market orders that were at least large enough to match limit orders (of our agent) placed at the best ask; these results were randomly generated for the simulation shown in Figure 1 (EDF stock). The gray bars represent the part that is filled by limit orders of our agent, while the green and red bars capture the parts that are filled with limit orders with the same, respectively with lower (more competitive) limit prices.

Figure 2 details the stochastic size and arrival time of the market orders that are depicted in the scenario for EDF stock in Figure 1. We observe that some market orders are executed against more aggressive limit orders during the last, say, 10 minutes of the 1-hour period (as shown by the red bars). This is because there are only a few shares left 10 minutes before the end of the period which gives the

²¹Kratz and Schöneborn [2012] find a similar result in the setting where the additional liquidity is provided by a dark pool.

agent some time to try to get a higher price for his (remaining) shares by placing limit orders away from the best ask price, accepting the fact that he might miss out on parts of the liquidity offered by market orders.

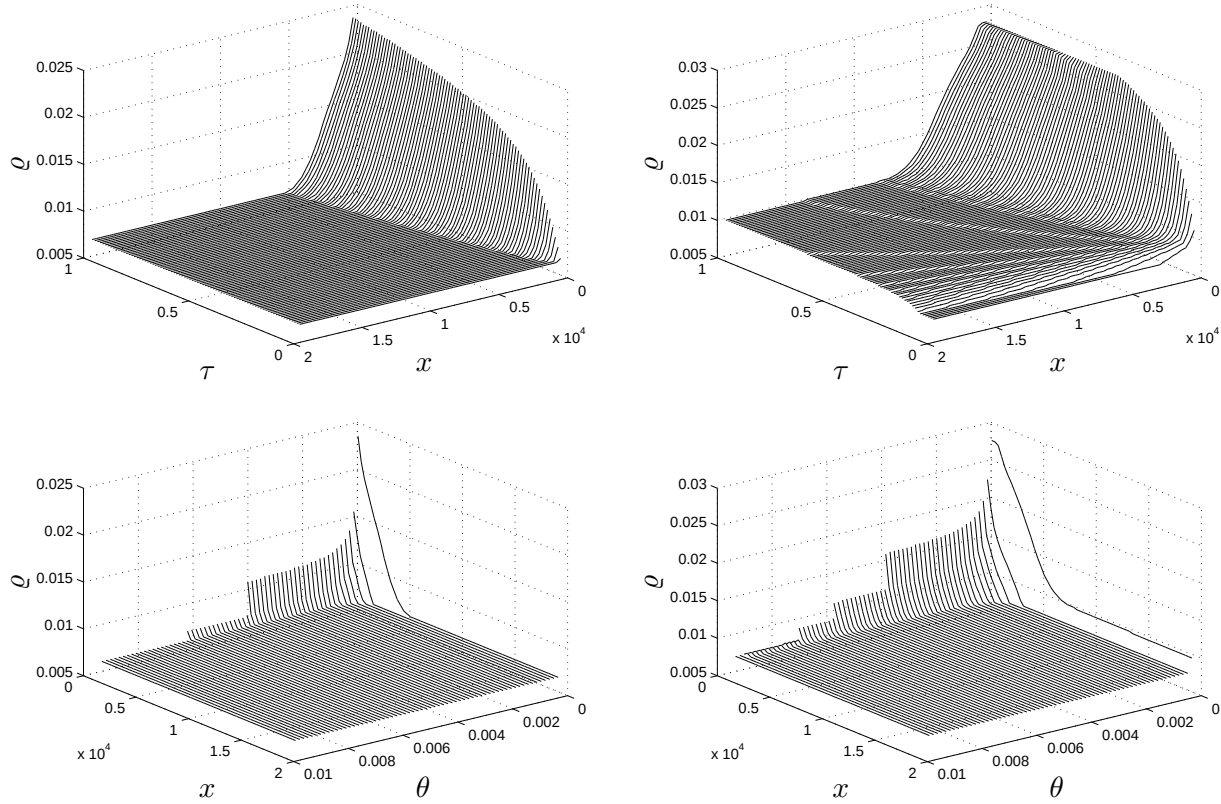


Figure 3: The optimal limit price ρ for the strategy on EDF stock (left graphs) and GD stock (right graphs) as a function of τ and x (upper graphs, $\theta = 10^{-4}$) and x and θ (lower graph, $\tau = 1$).

Figure 3 illustrates the following three points. First, as we also just saw in Figure 2, if the asset position x is small relative to the time τ left for executing it, then it is beneficial to post limit orders slightly away from the best price, but not by more than (roughly) 2 cents, i.e., 4 ticks (in case of EDF stock). Intuitively this makes sense since in this case agents are less restricted by time and can afford to wait for their limit orders to be filled. Esser and Monch [2007], Guéant et al. [2012] and Bayraktar and Ludkovski [2012] report a similar relationship between the optimal limit price and both the asset position x and the remaining time τ . Second, if x is large relative to τ , then it is beneficial to place limit orders at the best quote, which is here depicted by the level 6.75×10^{-3} for EDF stock and 7.85×10^{-3} for GD stock (half the size of their bid-ask spreads). Third, the optimal limit price is lower for high values of θ . The reason for this result is that transaction risk increases with the value of θ .

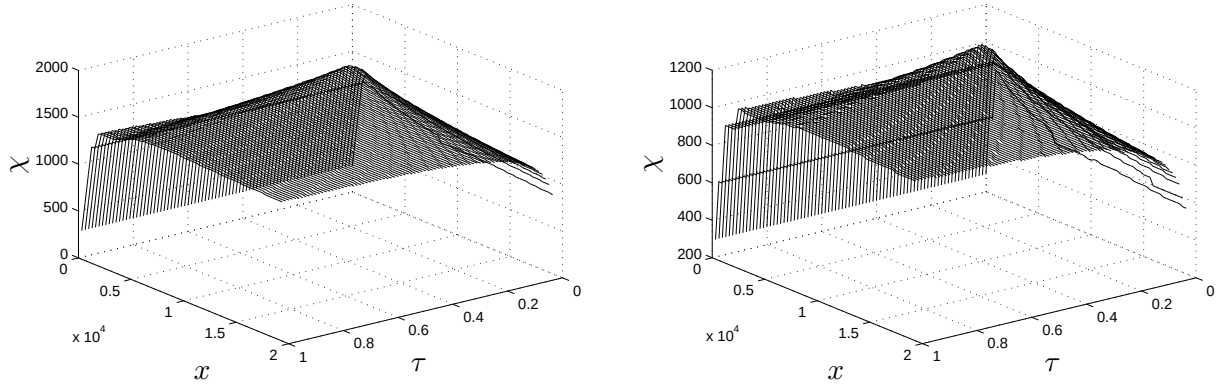


Figure 4: The optimal limit size χ as a function of τ and x ($\theta = 10^{-4}$). The left graph is for EDF stock, the right graph for GD stock.

In Figure 4 we have plotted how the optimal limit size varies with τ and x . The optimal χ is equal to x for x small and then flattens and slightly decreases for x large. This means that it is optimal to submit the entire risky-asset position x as a limit order when it is not too large and to submit (only) a fraction when it gets larger. The explanation for χ being decreasing for x large is that v is then large too, which means that the agent is more affected by the price impact of his limit orders. This effect is more pronounced when the time for execution is shorter. Finally, we observe that the optimal limit price is higher for EDF stock than for GD stock, which is due to the fact that order flow has (relatively) a higher impact on the price of GD shares than on the price of EDF shares.

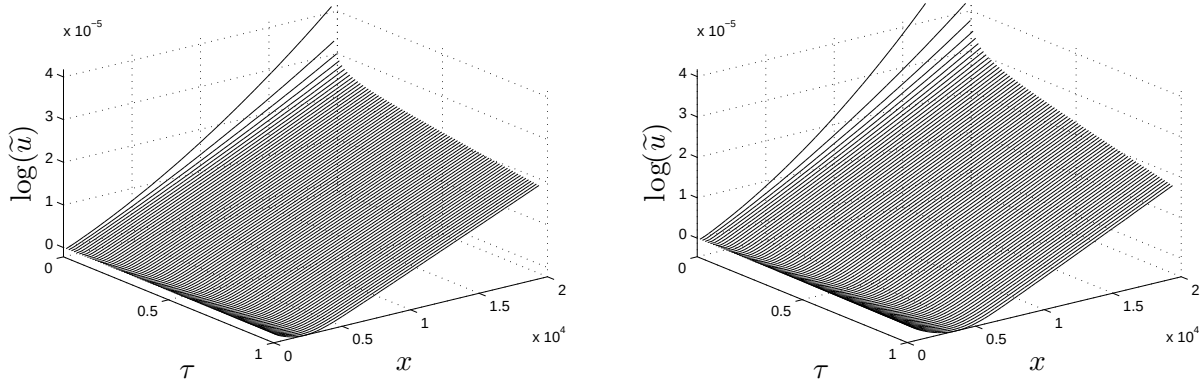


Figure 5: The logarithmic value function as a function of τ and x ($\theta = 10^{-4}$). The left graph is for EDF stock, the right graph for GD stock.

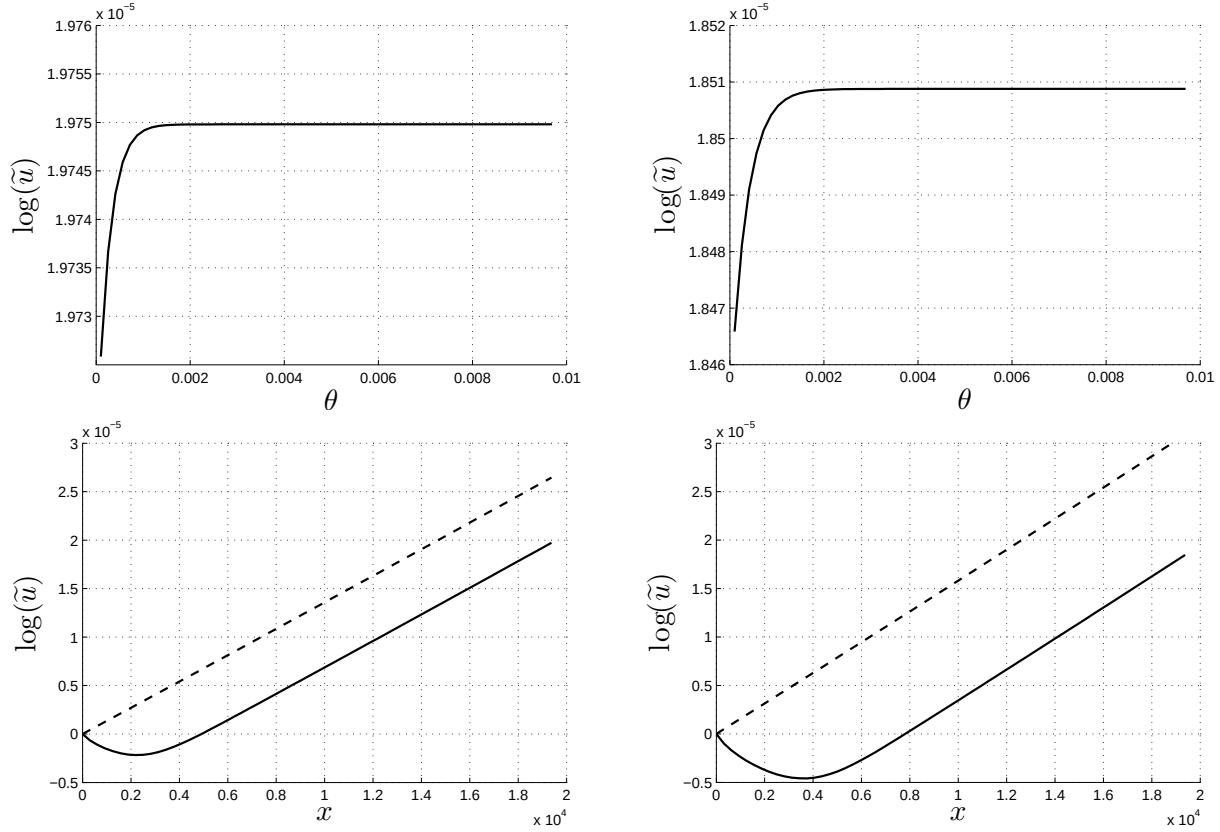


Figure 6: The logarithmic value function as a function of θ (upper graphs, $x = 2 \times 10^4$, $\tau = 1$) and x (lower graphs, $\theta = 10^{-4}$, $\tau = 1$). The dotted line corresponds to the logarithmic value function of a pure market order strategy. The left graphs are for EDF stock, the right graphs for GD stock.

Figures 5-6 show the logarithmic value function as a function of τ , x and θ . We recall that the value function stands for the loss in expected utility due to the cost of trading. We can draw several conclusions from these figures. First, we observe that the value function increases with θ . This effect arises from the negative correlation between the likelihood of limit orders being filled and the time since the arrival of the last market order. Second, the high values of the value function around $\tau = 0$ reflect the costly task of executing shares in a short period of time. Third, as opposed to a pure market order setting, the value function is not strictly increasing in x . The explanation is that shares are more likely to be executed as limit orders when x is small relative to τ . Hence the inclusion of limit orders reverts the slope of the value function for small x . Last, the value function is lower (higher) for a position in EDF stock than for a position in GD stock when x is large (small) relative to τ . The reason why it is lower when x is large relative to τ is because the price impact of market orders (as measured by β) and order flow (as measured by φ) is much larger for GD stock. The reason why it is higher when x is small relative to τ is because the bid-ask spread of EDF stock is narrower than the bid-ask spread of GD stock which leads to a higher value function when shares are more likely to be executed as a limit order than as a market order (which is the case when x is small relative to τ).

Figure 7 plots the optimal submission rate of market orders as a function of x and θ (τ is fixed at 1). We draw the following conclusions: First, the speed of trading increases with x and is significantly lower for a strategy involving limit orders. Moreover, for small values of x it is optimal to submit no market orders at all and to rely entirely on limit orders. This again reflects the fact that agents are willing to incur some market risk to potentially obtain a better price when someone fills their limit orders. Second, the speed of trading also increases with θ . This is due to the fact that limit orders are less likely to be filled

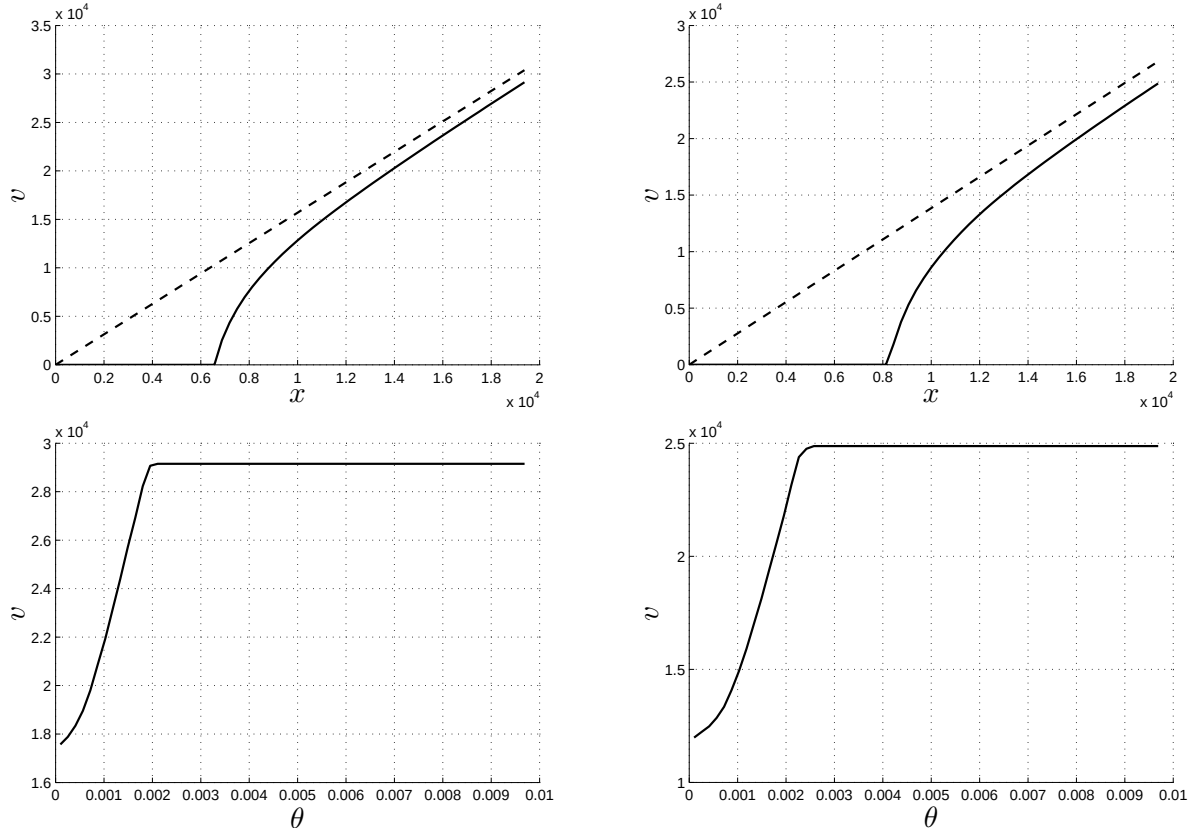


Figure 7: The optimal speed of trading as a function of x (upper graphs, $\tau = 1$, $\theta = 10^{-4}$) and θ (lower graphs, $\tau = 1$, $x = 2 \times 10^4$). The left graphs are for EDF stock, the right graphs for GD stock.

rapidly when θ is large. In this case the agent compensates by complementing his limit order strategy with more market orders.

5 Conclusions

In this paper we present a model that enables the analysis of trade execution strategies involving market and limit orders. We take the perspective of a risk-averse agent facing the problem of executing a large risky asset position within a finite time period. This problem requires balancing the following trade-off: If the agent trades quickly he is likely to incur higher costs due to price impact, more gradual trading brings about a larger exposure to price fluctuations, while a greater reliance on limit orders increases the uncertainty of execution. The agent solves this trade-off in an optimal fashion by dynamically adjusting the submission rate of market orders and the size and price of his limit orders, such that the expected exponential utility of his final cash position is maximized.

We assume that the price impact is a function of the order size over the lag variable that tracks the time since the last trade. In order to model the liquidity available for executing limit orders we assume the existence of other agents who randomly submit market orders. We further assume that these orders arrive according to a Poisson process.

We obtain the optimal trading strategy via a HJB equation that we derive using dynamic programming. We solve this first-order PIDE using the ADI finite-difference technique. The inversion of the non-sparse matrix describing the integral operator is circumvented by using a process in which matrix inversion is replaced by Fourier transforms.

We have tested our model on a set of real data and found that the possibility to trade limit orders reduces and sometimes even eliminates the use of market orders. Since limit orders are being executed at a higher price agents are willing to wait for them to be filled before increasingly complementing them with market orders. This effect is strongest for agents with a low degree of risk aversion, since they care less about the risk of fluctuating prices and uncertain execution.

In addition to this, we show that limit orders should be posted away from the market price when any of the following hold: i) trade execution is not urgent, ii) the asset position is relatively small, iii) the agent is only mildly risk averse. Finally, we show that the optimal size of limit orders is largest for moderate portfolio sizes and long time horizons.

Building on our results there are several interesting research questions that might prove worthwhile investigating. We believe that an interesting extension of this work would be to study the case where there are two or more dependent assets. A further extension would be to calibrate the model against a larger data set. Finally, another direction would be to research what happens when one models adverse selection by assuming that the price process and the liquidity for executing market and limit orders, are correlated.

A Proofs of the Propositions

Proposition 1. *Assume that there are other agents submitting market buy (sell) orders over any time interval $(s, t]$ ($0 \leq s < t \leq T$) with positive probability. Then the expected price impact cost of a market buy (sell) order can be reduced by breaking up the order into smaller pieces.*

Proof. Submitting a market buy order of size $\zeta > 0$ at time $t \in [0, T]$ has a cost of

$$c := \zeta \psi \left(\frac{\zeta}{\theta(t-)} \right). \quad (26)$$

Splitting the order into two sub-orders of equal size (without loss of generality), and then placing them in the market at time $t^* \in [0, t]$ and at time t , respectively, gives a total cost of

$$c^* := \frac{1}{2} \zeta \psi \left(\frac{\zeta}{2\theta(t^*)} \right) + \frac{1}{2} \zeta \psi \left(\frac{\zeta}{2 \min\{\theta(t-), t - t^*\}} \right), \quad (27)$$

where θ is the trade-lag without taking the trade at time t^* into account. Now take t^* equal to $\min\{s \in [0, t] \mid \theta(s-) = t - s\}$. This minimum always exists since $\theta(0) = 0 \leq t$. The definition of t^* gives $\theta(t) \leq 2(t - t^*)$, or, equivalently, $c \geq c^*$, where the inequality is strict if, and only if, a market buy order arrives in the interval $(t^*, t]$. Since the latter occurs with positive probability (by assumption), the proposition follows. **q.e.d.**

Proposition 2. *The price impact of an admissible continuous-time market sell order strategy $\{v(t)\}_{t \in [0, T]}$ at time $t \in [0, T]$ is given by*

$$\psi(-v(t)).$$

Proof. Consider the following discrete-time approximation of the strategy $\{v(t)\}_{t \in [0, T]}$:

$$\left\{ \left(t_i, v(t_i) \frac{T}{m} \right) \right\}_{i=1, \dots, m}, \quad (28)$$

where $t_i = iT/m$ and $m \in \mathbb{N}_{>0}$. The total cost of this strategy over the time interval $[0, t]$ can be written as

$$-\frac{T}{m} \sum_{i=1}^{m^*} v(t_i) \psi \left(-\frac{v(t_i)T}{\theta(t_i-)m} \right), \quad (29)$$

where $m^* = \max\{i \leq m \mid t_i \leq t\}$ and $\theta(t_i-) = \frac{T}{m} - (\nu(t_{i-1}) - t_{i-1}) \mathbf{1}_{\{\nu(t_{i-1}) < t_i\}}$. (Recall that $\nu(t)$ is the first time after time $t \in [0, T]$ that a market buy order arrives.) The probability that $\nu(t_{i-1}) < t_i$ is proportional to $\frac{T}{m}$ for large m , which is a property of the Poisson process, and thus we have

$$\lim_{m \rightarrow \infty} \theta(t_i-)m = T.$$

Since $\xi \mapsto \xi \psi(\xi)$ is continuous on $\mathbb{R}$ and v is Riemann integrable (because of its bounded variation), $v(t) \psi(v(t))$ is Riemann integrable, too. Hence we find that the price impact of $\{v(t)\}_{t \in [0, T]}$ over the period $[0, t]$ is given by

$$\lim_{m \rightarrow \infty} -\frac{T}{m} \sum_{i=1}^{m^*} v(t_i) \psi(-v(t_i)) = -\int_0^t v(s) \psi(-v(s)) ds, \quad (30)$$

for all $t \in [0, T]$. Differentiating the RHS above by t and dividing by $-v(t)$ proves the proposition. **q.e.d.**

Proposition 3. *The minimization problem in (14) admits a solution.*

Proof. We write the minimization function as

$$\begin{aligned} M : (v, \chi, \varrho) &\mapsto -v\tilde{u}_x - \gamma(\psi(-v) - \varphi\chi)v\tilde{u} \\ &+ \lambda \int_0^\infty [\tilde{u}(\tau, x - q \wedge \chi, 0) \exp(-\gamma(\varrho - \varphi\chi)(q \wedge \chi)) - \tilde{u}_0] \exp(-\gamma\varphi x(q + \theta\Psi^{-1}(\varrho))) f_Q(q + \theta\Psi^{-1}(\varrho)) dq. \end{aligned}$$

Since, by assumption, ψ , Ψ^{-1} and f_Q are continuous on $[0, \infty]$, $[\alpha, \infty]$, respectively on $[0, \infty]$, we have that M is a continuous function. Furthermore, since $\tilde{u} > 0$ and since $v \mapsto -\psi(-v)v$ is strictly convex (because of condition (ii) of the PIF), M is strictly convex with respect to v . This means $v^*(\chi)$ (whose dependence on χ is made explicit) always exists. Now assume that $M(v, \chi, \varrho) = \varepsilon < M(v^*(0), 0, 0)$ for some $(v, \chi, \varrho) \in A$ (otherwise, $(v, \chi, \varrho) = (v^*(0), 0, 0)$ is a solution). Under this assumption, we obtain the following result:

$$\lim_{\varrho \rightarrow \infty} M(v, \chi, \varrho) \geq M(v, 0, 0) > \varepsilon \quad \text{for all } v \geq 0 \text{ and } 0 \leq \chi \leq x.$$

Consider the set $A_\varepsilon := \{(v, \chi, \varrho) \in A \mid M(v, \chi, \varrho) \leq \varepsilon\}$. From the above results, this set is non-empty, closed and bounded (i.e. compact), and hence, by Weierstrass' Theorem, it has a global minimizer, which is then also a global minimizer of M on the set A . **q.e.d.**

Proposition 4. *A solution to the minimization problem in (14) with $\psi(\xi) = \alpha + \beta\xi$ and $f_Q(q) = \frac{\kappa}{2} \exp(-\kappa|q|)$ is unique on a subspace of A (which is defined in the proof).*

Proof. Substituting $\psi(\xi) = \alpha + \beta\xi$ and $f_Q(q) = \frac{\kappa}{2} \exp(-\kappa|q|)$ into the function M (as defined in the proof of Proposition 3) gives

$$M : (v, \chi, \varrho) \mapsto -v\tilde{u}_x + \gamma(\alpha + \beta v + \varphi\chi)v\tilde{u} + \frac{\lambda\kappa}{2} \int_0^\infty [\exp(\omega(q \wedge \chi, \varrho - \varphi\chi)) - \tilde{u}_0] \exp(-\varsigma(2\beta q + \theta(\varrho - \alpha))) dq,$$

where $\omega : (q, \varrho) \mapsto \log \tilde{u}(\tau, x - q, 0) - \gamma \varrho q$ and $\varsigma = (\kappa + \gamma \varphi x)/(2\beta)$.

Let B denote a subspace of A which will be defined later. We first prove that a global minimizer of M on $\mathring{B}$ (interior of the set B) is unique on B by showing that $H_{v,\chi,\varrho}(M)(v, \chi, \varrho)$ is positive definite at every point $(v, \chi, \varrho) \in \mathring{B}$ satisfying $\nabla_{v,\chi,\varrho}(M)(v, \chi, \varrho) = 0$ (where ∇_x and H_x denote the gradient and Hessian with respect to the variable(s) x , respectively). Then, by Theorem 3.2 in Demidenko [2008], the point $(v, \chi, \varrho) \in \mathring{B}$ satisfying $\nabla_{v,\chi,\varrho} M(v, \chi, \varrho) = 0$ is a unique global minimizer on B . Subsequently we show along the same lines that a global minimizer of M on $B \setminus \mathring{B}$ is unique on $B \setminus \mathring{B}$. Then, by using that a global minimizer of M on $\mathring{B}$ is unique on B , we have that a unique global minimizer of M on $B \setminus \mathring{B}$ is also a unique global minimizer of M on B .

Positive definiteness of the Hessian matrix for M is equivalent to the inequality

$$M_{vv}(v, \chi, \varrho)[M_{\chi\chi}(v, \chi, \varrho)M_{\varrho\varrho}(v, \chi, \varrho) - M_{\chi\varrho}^2(v, \chi, \varrho)] - M_{v\chi}^2(v, \chi, \varrho)M_{\varrho\varrho}(v, \chi, \varrho) > 0, \quad (31)$$

where we have used that $M_{v\varrho}(v, \chi, \varrho) = 0$. First, we note that $M_{vv}(v, \chi, \varrho) = 2\gamma\beta\tilde{u}$ is positive on A (and thus also on B). Then we differentiate M once and twice with respect to ϱ and χ , to find

$$\begin{aligned} M_{\varrho}(v, \chi, \varrho) &= -\frac{\lambda\kappa\gamma}{2} \int_0^\infty (q \wedge \chi) \exp(\omega(q \wedge \chi, \varrho - \varphi\chi) - \varsigma(2\beta q + \theta(\varrho - \alpha))) dq - \varsigma\theta m_1(\chi, \varrho), \\ M_{\chi}(v, \chi, \varrho) &= \gamma\varphi v\tilde{u} + \omega_q(\chi, \varrho - \varphi\chi)m_2(\chi, \varrho) + \frac{\lambda\kappa\gamma\varphi}{2} \int_0^\infty (q \wedge \chi) \exp(\omega(q \wedge \chi, \varrho - \varphi\chi) - \varsigma(2\beta q + \theta(\varrho - \alpha))) dq, \\ M_{\varrho\varrho}(v, \chi, \varrho) &= \frac{\lambda\kappa\gamma}{2} \int_0^\infty (q \wedge \chi) [(q \wedge \chi)\gamma + \varsigma\theta] \exp(\omega(q \wedge \chi, \varrho - \varphi\chi) - \varsigma(2\beta q + \theta(\varrho - \alpha))) dq - \varsigma\theta M_{\varrho}(v, \chi, \varrho), \\ M_{\chi\chi}(v, \chi, \varrho) &= [\omega_q(\chi, \varrho - \varphi\chi)(\omega_q(\chi, \varrho - \varphi\chi) - \varsigma(2\beta + \theta\varphi)) + \omega_{qq}(\chi, \varrho - \varphi\chi)]m_2(\chi, \varrho) - 2\varphi M_{\chi\varrho}(v, \chi, \varrho) \\ &\quad - \varphi^2 M_{\varrho\varrho}(v, \chi, \varrho) - \varsigma\theta\varphi[M_{\chi}(v, \chi, \varrho) - \varphi M_{\varrho}(v, \chi, \varrho) - \gamma\varphi v\tilde{u}] \\ M_{\chi\varrho}(v, \chi, \varrho) &= -[\omega_q(\chi, \varrho - \varphi\chi)(\gamma\chi + \varsigma\theta) + \gamma]m_2(\chi, \varrho) - \varphi M_{\varrho\varrho}(v, \chi, \varrho) - \varsigma\theta\varphi M_{\varrho}(v, \chi, \varrho), \end{aligned}$$

where the functions $m_1 : [0, x] \times [\alpha, \infty] \rightarrow [0, \infty]$ and $m_2 : [0, x] \times [\alpha, \infty] \rightarrow [0, \infty]$ are defined by

$$m_1(\chi, \varrho) := \frac{\lambda\kappa}{2} \int_0^\infty [\exp(\omega(q \wedge \chi, \varrho - \varphi\chi)) - \tilde{u}_0] \exp(-\varsigma(2\beta q + \theta(\varrho - \alpha))) dq,$$

respectively, by

$$m_2(\chi, \varrho) := \frac{\lambda\kappa}{2} \int_\chi^\infty \exp(\omega(\chi, \varrho - \varphi\chi) - \varsigma(2\beta q + \theta(\varrho - \alpha))) dq \geq 0.$$

From the above expressions we obtain $M_{\chi\chi}(v, \chi, \varrho) > 0$ and $M_{\varrho\varrho}(v, \chi, \varrho) > 0$. Furthermore, by using that $M_{\chi}(v, \chi, \varrho) = 0$ at points where the gradient vanishes, we also obtain

$$-\omega_q m_2 \geq \gamma\varphi\chi m_2 + \gamma\varphi v\tilde{u}, \quad (32)$$

where, from now on, we have suppressed arguments for notational ease. Putting together the expressions for the partial derivatives of M , and by using that $M_{\varrho}(v, \chi, \varrho) = M_{\chi}(v, \chi, \varrho) = 0$, we find

$$\begin{aligned} M_{\varrho\varrho} &> -\varsigma\theta\gamma v\tilde{u} + \left(\gamma^2\chi^2 - \frac{\varsigma\theta}{\varphi}\omega_q\right)m_2, \\ M_{\chi\chi} &> \omega_q[\omega_q - \varsigma(2\beta + \theta\varphi)]m_2 - 2\varphi M_{\chi\varrho} - \varphi^2 M_{\varrho\varrho} + \varsigma\theta\varphi^2\gamma v\tilde{u}, \\ M_{\chi\varrho}^2 &= [\omega_q(\gamma\chi + \varsigma\theta) + \gamma]^2 m_2^2 - 2\varphi M_{\varrho\varrho}M_{\chi\varrho} - \varphi^2 M_{\varrho\varrho}^2, \end{aligned}$$

where we have used that $\omega_{qq} > 0$.²² Now we assume that the following conditions are satisfied:

$$(i) \ v \geq \frac{\varphi}{2\varsigma\beta(\theta\varphi + 2\beta)}, \quad (ii) \ \chi > \max \left\{ -\frac{1}{\varphi\varsigma\theta} - \frac{\omega_q}{\gamma\varphi}, -\frac{\varsigma\theta}{\gamma} - \frac{1}{\omega_q}, 0 \right\}.$$

We point out that condition (ii) implies that $\gamma < -\omega_q(\gamma\chi + \varsigma\theta)$ and $-\varsigma\theta(\omega_q + \gamma\varphi\chi) > \gamma$ which together with (32) gives $\varsigma\theta\varphi v\tilde{u} < m_2$. Under conditions (i) and (ii), which define the subspace B of A , and using (32), we obtain the following result:

$$\begin{aligned} & [M_{\chi\chi} - M_{v\chi}^2/M_{vv}]M_{\varrho\varrho} - M_{\chi\varrho}^2 \\ & > \left[\omega_q(\omega_q - \varsigma(2\beta + \theta\varphi))m_2 + \varsigma\theta\varphi^2\gamma v\tilde{u} - \frac{\gamma\varphi^2\tilde{u}}{2\beta} \right] \left[-\varsigma\theta\gamma v\tilde{u} + \left(\gamma^2\chi^2 - \frac{\varsigma\theta}{\varphi}\omega_q \right) m_2 \right] - [\omega_q(\gamma\chi + \varsigma\theta) + \gamma]^2 m_2^2 \\ & > \omega_q(\omega_q - \varsigma\theta\varphi)m_2 \left[-\varsigma\theta\gamma v\tilde{u} + \left(\gamma^2\chi^2 - \frac{\varsigma\theta}{\varphi}\omega_q \right) m_2 \right] - [\omega_q(\gamma\chi + \varsigma\theta) + \gamma]^2 m_2^2 \\ & = \varsigma\theta\gamma v\tilde{u}(\varsigma\theta\varphi - \omega_q)\omega_q m_2 - \frac{\varsigma\theta}{\varphi}(\gamma\varphi\chi + \omega_q)^2 \omega_q m_2^2 - 2\gamma \left[\omega_q(\gamma\chi + \varsigma\theta) + \frac{\gamma}{2} \right] m_2^2 \\ & = \varsigma\theta\gamma v\tilde{u}(\varsigma\theta\varphi - \omega_q)\omega_q m_2 + \varsigma\theta\gamma v\tilde{u}(\gamma\varphi\chi + \omega_q)\omega_q m_2 - \gamma(\gamma\chi + \varsigma\theta)\omega_q m_2^2 \\ & = \gamma(\varsigma\theta\varphi v\tilde{u} - m_2)(\gamma\chi + \varsigma\theta)\omega_q m_2 \\ & > 0, \end{aligned}$$

which shows that (31) holds on B . From the above it also follows that the Hessian matrix of M with respect to v and χ is positive definite at the boundary $\varrho = \alpha$, and so is the Hessian matrix of M with respect to χ and ϱ at the boundary $v = \text{const.} \geq 0$ except when $\chi = 0$, but points where $\chi = 0$ do not belong to the subspace B (by condition (ii) above). Furthermore, since $M_{v\varrho} = 0$, the Hessian matrix of M with respect to v and ϱ is positive definite at the boundary $\chi = x$. This concludes the proof. **q.e.d.**

B Derivation of the HJB Equation

Taking the (partial) derivative of time on the RHS of (12) gives

$$\begin{aligned} & v\hat{u}_x - \frac{\hat{u}_\theta}{2} - \left[\frac{\gamma^2\sigma^2}{2}x^2 - \gamma(\psi(-v) - \varphi\chi)v \right] \hat{u} + \lambda\hat{u} \\ & - \lambda \int_{-\infty}^{\infty} \left(\hat{u}(t, x - ((q - \theta\Psi^{-1}(\varrho))_+ \wedge \chi), \theta\mathbf{1}_{\{q \leq 0\}}) \exp(-\gamma\varphi xq - \gamma(\varrho - \varphi\chi)((q - \theta\Psi^{-1}(\varrho))_+ \wedge \chi)) \right) f_Q(q) dq, \end{aligned}$$

which can be rewritten as

$$\begin{aligned} & v\hat{u}_x - \frac{\hat{u}_\theta}{2} - \left[\frac{\gamma^2\sigma^2}{2}x^2 - \gamma(\psi(-v) - \varphi\chi)v \right] \hat{u} + \lambda \left[(1 - F_Q^e(x, 0))\hat{u} - (F_Q^e(x, \infty) - F_Q^e(x, 0))\hat{u}_0 \right] \\ & - \lambda \int_0^\infty \left[\hat{u}(t, x - q \wedge \chi, 0) \exp(-\gamma(\varrho - \varphi\chi)(q \wedge \chi)) - \hat{u}_0 \right] \exp(-\gamma\varphi x(q + \theta\Psi^{-1}(\varrho))) f_Q(q + \theta\Psi^{-1}(\varrho)) dq \end{aligned}$$

²²This is easily proved by showing that $x \mapsto \tilde{u}(\tau, x, 0)$ is log-convex. Write $x = w\bar{x}_0 + (1-w)\bar{x}_1$ (with $w \in (0, 1)$ and $\bar{x}_0 < x < \bar{x}_1$). Then, one can verify that $\tilde{u}(\tau, x, 0) \leq \tilde{u}(\tau, w\bar{x}_0, 0)\tilde{u}((1-w)\tau, (1-w)\bar{x}_1, 0) \leq \tilde{u}(\tau, w\bar{x}_0, 0)\tilde{u}(\tau, (1-w)\bar{x}_1, 0)$, and hence, $x \mapsto \tilde{u}(\tau, x, 0)$ is log-convex.

By noting that the (partial) time derivative on the LHS of (12) ought to be equal to the above expression if the optimal control is chosen over the interval $(t, t + dt)$, we obtain

$$0 = \hat{u}_t + \min_{(v, \chi, \varrho) \in A} \left\{ -v\hat{u}_x + \frac{\hat{u}_\vartheta}{2} + \left[\frac{\gamma^2 \sigma^2}{2} x^2 - \gamma(\psi(-v) - \varphi\chi)v \right] \hat{u} - \lambda \left[(1 - F_Q^e(x, 0))\hat{u} - (F_Q^e(x, \infty) - F_Q^e(x, 0))\hat{u}_0 \right] + \lambda \int_0^\infty [\hat{u}(t, x - q \wedge \chi, 0) \exp(-\gamma(\varrho - \varphi\chi)(q \wedge \chi)) - \hat{u}_0] \exp(-\gamma\varphi x(q + \theta\Psi^{-1}(\varrho))) f_Q(q + \theta\Psi^{-1}(\varrho)) dq \right\},$$

which is the HJB equation.

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